VORTICITY-ASSISTED MIXING AND SHARP OPENING RATES FOR THE MUSKAT PROBLEM WITH VISCOSITY JUMP

DRAFT MANUSCRIPT

Abstract. We study mixing solutions of the incompressible porous media equation for two fluids with different densities and different viscosities. Luna-Velasco and Mengual recently constructed such solutions for *fully unstable* Muskat data, i.e. when the Saffman–Taylor function σ_0 of the initial interface is negative everywhere. We show that the second order compatibility condition in their time expansion contains a free function, which can be used to *tilt* the relaxed flux inside the mixing zone towards the centre of the relaxation disc. This yields a sharp description of single-strip mixing zones: a strip with intermediate density ρ_{m} and opening function κ emanating from an interface z_0 exists only if its centre of mass starts with the Muskat velocity and

$$\kappa\big(2(1-A\rho_{\mathrm{m}})\kappa+\sigma_0\big)\le\frac{(1-A^2)\,\gamma_0^2}{8(1+A\rho_{\mathrm{m}})},$$

and it exists whenever the strict inequality holds. Here A is the viscosity Atwood number and γ_0 is the strength of the vortex sheet carried by the interface. Consequently, mixing solutions exist for every initial interface of class $C^{6,\gamma}$ outside the *pinch set* $\{\sigma_0\ge 0,\ \gamma_0=0\}$, including interfaces that are partially Saffman–Taylor *stable*, where mixing is driven by the vortex sheet; for equal viscosities this covers every overturned interface whose tangent never points backwards. The results also apply, after a change of frame, to viscous fingering driven by injection without any density contrast. The analysis rests on writing the Darcy field $\mu u=-\nabla p-\rho e_2$ of a subsolution as Cauchy integrals whose densities are linear in the Saffman–Taylor function, and on the invertibility of $I\mp 2A\mathcal{K}$, where $\mathcal{K}$ is the adjoint Neumann–Poincaré operator of the interface.

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1. Introduction

1.1. **The equations.** We study two incompressible fluids in a two-dimensional, horizontally periodic porous medium (or Hele-Shaw cell) under gravity. The fluids have different densities

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Provenance and credit. The existence of mixing solutions for *fully unstable* Muskat data with viscosity jump, the Saffman–Taylor formulation of the Muskat problem and the single-layer construction with a second order expansion in time are due to P. Luna-Velasco and F. Mengual [LM26]. Sections 2–4 contain a self-contained (horizontally periodic, Hölder) account of that framework, written independently from the abstract of [LM26] before the paper was read; see `COMPARISON.md` in the source repository. The new contributions are Theorems 1.1 and 1.2, Corollaries 1.3–1.5 and the tilting mechanism of Section 5; their novelty with respect to [CFM22] for equal viscosities has not been checked against the full text of that paper.

and different viscosities, and their evolution is governed by the *incompressible porous media equation with viscosity jump*

$$\partial_t\rho + \operatorname{div}(\rho u) = 0, \qquad \operatorname{div} u = 0, \qquad \mu(\rho)\, u = -\nabla p - \rho\, e_2, \qquad (x,t) \in \mathbb{T}\times\mathbb{R}\times(0,T), \tag{1.1}$$

where $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, $e_2 = (0,1)$, u is the velocity, p the pressure, ρ the density and $\mu(\rho)$ the viscosity (permeability and gravity are normalised to one). After subtracting a hydrostatic pressure and rescaling,

$$\rho \in \{-1,1\}, \qquad \mu(\rho) = 1 + A\rho, \qquad A = \frac{\mu^+ - \mu^-}{\mu^+ + \mu^-} \in (-1,1), \tag{1.2}$$

where $\mu^\pm = 1 \pm A$ are the viscosities of the heavy ($\rho = 1$) and light ($\rho = -1$) fluids and A is the viscosity Atwood number. Since μ is affine in ρ and $\rho^2 = 1$, Darcy's law can be written linearly as $u + Am = -\nabla p - \rho e_2$ with $m = \rho u$ [Men22].

When $\rho = 1$ above a curve $\Gamma(t)$ and $\rho = -1$ below it, (1.1) is the *Muskat problem* [Mus34]. Two scalar quantities on the interface govern it: the *Saffman–Taylor function* and the *vortex-sheet strength*

$$\sigma := \partial_n p^+ - \partial_n p^-, \qquad \gamma := u^+\cdot\tau - u^-\cdot\tau, \tag{1.3}$$

where n and τ are the upward normal and the tangent, and $\pm$ denote traces from above and below. If $\sigma > 0$ the problem is locally well-posed [Amb04, CG07, CCG11, Mat19, NP20] (see the surveys [Gan17, GBL20]); if $\sigma < 0$ it is ill-posed [SCH04, CG07]: this is the Saffman–Taylor instability [ST58, Hom87]. In our normalisation $\sigma = -2(n_2 + AU)$, where $U = u\cdot n$, so the instability has a gravitational part ($n_2 > 0$: heavy fluid on top) and a viscous part ($AU > 0$: the less viscous fluid pushes into the more viscous one).

1.2. **Mixing solutions.** In the unstable regime one looks for weak solutions in which the fluids mix at small scales inside a growing *mixing zone*. Otto [Ott99] described the flat case by a relaxation based on the gradient-flow structure; Székelyhidi [Sze12], building on [DLS09, CFG11, Shv11], computed the relaxation of IPM for $A = 0$ and constructed mixing solutions by convex integration. For curved interfaces and $A = 0$, mixing solutions were constructed for fully unstable data in [CCF21] (see also [ACF21]), by piecewise constant subsolutions in [FS18, NS21], with degraded profiles in [CFM19], and with mixing zones localised near the unstable part of bubbles and turned interfaces in [CFM22]. For $|A| < 1$, Mengual [Men22] proved the h-principle and treated the flat interface, observing that a *pinch singularity* of the relaxation prevents mixing wherever there is neither Rayleigh–Taylor instability nor vorticity.

Very recently, Luna-Velasco and Mengual [LM26] constructed mixing solutions for curved interfaces with viscosity jump, under the assumption that the data are *fully unstable*, $\sigma_0 < 0$ everywhere. Their key idea is to formulate the inversion of the Birkhoff–Rott operator in terms of σ rather than the vorticity. Their subsolutions consist of one (or finitely many) strips of constant intermediate density around a central curve, defined through a second order expansion in time as in [FS18, CFM22]; the opening speeds are proportional to σ_0.

1.3. **Main results.** Let $z_0 \in \mathcal{Z}^{k,\gamma}$, $k \geq 3$, be a horizontally periodic interface (Definition 2.1) with upward normal n_0 and tangent τ_0. By Propositions 2.7 and 2.10, the Saffman–Taylor function, the normal velocity and the vortex-sheet strength of the Muskat problem at $t = 0$ are

$$\sigma_0 = -2\big(I - 2A\mathcal{K}_0\big)^{-1}[n_{0,2}], \qquad U_0 = -\mathcal{K}_0\sigma_0, \qquad \gamma_0 = -2\big(I + 2A\mathcal{K}_0\big)^{-1}[\tau_{0,2}], \tag{1.4}$$

where $\mathcal{K}_0$ is the adjoint Neumann–Poincaré operator of z_0. For an intermediate density $\rho_{\mathrm{m}} \in (-1,1)$ let $\mu_{\mathrm{m}} = 1 + A\rho_{\mathrm{m}}$ and define the *admissibility function* and the *maximal opening function*

$$\Phi(\kappa) := \kappa\big(2(1-A\rho_{\mathrm{m}})\kappa + \sigma_0\big) - \frac{(1-A^2)\gamma_0^2}{8\mu_{\mathrm{m}}}, \quad \kappa_{\max} := \frac{-\sigma_0 + \sqrt{\sigma_0^2 + (1-A\rho_{\mathrm{m}})(1-A^2)\gamma_0^2/\mu_{\mathrm{m}}}}{4(1-A\rho_{\mathrm{m}})}. \tag{1.5}$$

For $\kappa \geq 0$, $\Phi(\kappa) < 0$ if and only if $\kappa < \kappa_{\max}$. Finally, the *pinch set* of z_0 is

$$\Pi(z_0) := \{\alpha \in \mathbb{T} :\ \sigma_0(\alpha) \geq 0 \ \text{ and } \ \gamma_0(\alpha) = 0\} = \{\kappa_{\max} = 0\}.$$

Note that $\kappa_{\max} \geq -\sigma_0/(2(1-A\rho_{\mathrm{m}}))$, with equality where $\gamma_0 = 0$ and $\sigma_0 < 0$.

Theorem 1.1 (Existence with sharp opening rate)**.** *Let $A \in (-1,1)$, $\gamma \in (0,1)$, $z_0 \in \mathcal{Z}^{6,\gamma}$, $\rho_{\mathrm{m}} \in (-1,1)$, and let $\kappa \in C^{4,\gamma}(\mathbb{T})$ satisfy*

$$0 < \kappa(\alpha) < \kappa_{\max}(\alpha) \qquad \text{for every } \alpha \in \mathbb{T}. \tag{1.6}$$

Then there exist $T > 0$, $a \in C^{3,\gamma}(\mathbb{T})$ and a polynomial h with $|h(t)| \leq Ct^3$ such that, setting

$$Z(\alpha,t) = z_0(\alpha) + \Big(tU_0(\alpha) + \tfrac{t^2}{2}a(\alpha)\Big)n_0(\alpha) + h(t)e_2, \qquad z^{\pm}(\alpha,t) = Z(\alpha,t) \pm t(1\pm\rho_{\mathrm{m}})\kappa(\alpha)n_0(\alpha), \tag{1.7}$$

and denoting by $\Omega_{\mathrm{mix}}(t)$ the open strip between the curves $\Gamma^{\pm}(t) = z^{\pm}(\mathbb{T},t)$, there are infinitely many weak solutions (ρ,u) of (1.1) on $[0,T]$ with the Muskat initial datum $\rho_0 = \mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}$ such that

(i) *$\rho = 1$ above $\Gamma^+(t)$ and $\rho = -1$ below $\Gamma^-(t)$;*
(ii) *ρ mixes in Ω_{mix}: both phases have positive measure in every open subset of $\{(x,t) : 0 < t < T,\ x \in \Omega_{\mathrm{mix}}(t)\}$;*
(iii) *a sequence of such solutions converges weakly-$*$ to the density equal to 1, ρ_{m}, -1 above, inside and below the strip.*

The condition (1.6) is sharp within the natural class of single-strip subsolutions (Definition 6.1):

Theorem 1.2 (Necessary conditions)**.** *Let $z_0 \in \mathcal{Z}^{3,\gamma}$ and $\rho_{\mathrm{m}} \in (-1,1)$, and let $(\bar\rho,\bar u,\bar m)$ be a single-strip subsolution with intermediate density ρ_{m} emanating from z_0, whose boundary curves have initial normal velocities $W^{\pm}$. Then*

(a) *the centre of mass starts with the Muskat velocity: $\frac{1}{2}[(1-\rho_{\mathrm{m}})W^+ + (1+\rho_{\mathrm{m}})W^-] = U_0$; equivalently $W^{\pm} = U_0 \pm (1\pm\rho_{\mathrm{m}})\kappa$ with $\kappa = \frac{1}{2}(W^+ - W^-) \geq 0$;*
(b) *$\Phi(\kappa) \leq 0$ on $\mathbb{T}$; that is, $\kappa \leq \kappa_{\max}$, and in particular $\kappa = 0$ on $\Pi(z_0)$.*

The single-layer subsolutions of [LM26] (with $\rho_{\mathrm{m}} = A$) have the properties of Definition 6.1 in the asymptotically flat setting, and the proof of Theorem 1.2 applies to them verbatim. Combining the two theorems:

Corollary 1.3 (Pinch-free data)**.** *Let $z_0 \in \mathcal{Z}^{6,\gamma}$. A single-strip subsolution emanating from z_0 with everywhere positive opening exists if and only if $\Pi(z_0) = \emptyset$. In that case the conclusions of Theorem 1.1 hold with $\kappa = \theta\kappa_{\max}$ for every $\theta \in (0,1)$. In particular, mixing solutions exist for data that are* not *fully unstable: on the part of the interface where $\sigma_0 \geq 0$ the mixing is driven by the vortex sheet.*

Corollary 1.4 (Equal viscosities)**.** *Let $A = 0$ and $z_0 \in \mathcal{Z}^{6,\gamma}$. Then $\kappa_{\max} = \frac{1}{2}(1+n_{0,2})$ and $\Pi(z_0) = \{\alpha :\ \tau_0(\alpha) = -e_1\}$. Hence, if the unit tangent of z_0 never equals $-e_1$, mixing solutions exist with any opening $0 < \kappa < \frac{1}{2}(1+n_{0,2})$; this includes overturned interfaces, which are Rayleigh–Taylor stable on the overturned part.*

Corollary 1.5 (Viscous fingering)**.** *The results above apply, after a change of frame and of time scale, to the problem with a uniform injection velocity Ve_2 at infinity and a gravity constant $G \geq 0$, provided $G + AV > 0$. In particular, for fluids of equal density ($G = 0$), mixing solutions exist whenever the less viscous fluid is injected into the more viscous one ($AV > 0$) and the data are pinch-free; see Corollary 7.1.*

Figure 1 illustrates Corollary 1.3 on an overturned interface.

1.4. **The mechanism: tilting the relaxed flux.** The relaxation of (1.1) is described in Section 3 by a vector D (a normalised relaxed flux) lying in the open disc $|D-a| < |a|$, where $a = e_2 + \frac{A}{\mu}v$ is computed from the Darcy field $v = \mu u$. In a thin mixing zone, mass conservation and the Rankine–Hugoniot conditions fix the *normal* component of D, namely $D\cdot n = -\sigma_0 - 2(1-A\rho_{\mathrm{m}})\kappa$ at leading order. The *tangential* component of D, on the other hand, is the transverse derivative of a stream function across a strip of width $O(t)$, and it is

controlled by the $O(t)$ part of the flux mismatch between the two boundaries. That part is in turn governed by the normal acceleration of the central curve, which is the unknown of the second order compatibility condition. In [LM26] this freedom is used to make the tangential component vanish. We use it instead to place D on the line through the centre of the disc. Since

$$a \cdot n = -\tfrac{1}{2}\sigma_0 + O(\kappa), \qquad a \cdot \tau = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}},$$

the admissible range is then governed by $|a|$, which combines the Saffman–Taylor function and the vortex-sheet strength, instead of $a \cdot n$ alone. This gives (1.6). Theorem 1.2 shows that nothing better can be achieved with a single strip: the normal component of D on the boundary of the strip is forced, and the hull condition at $t = 0$ is exactly $\Phi(\kappa) \le 0$.

1.5. **Relation to** [LM26]**.** Sections 2–4 give a self-contained account, in a horizontally periodic Hölder setting, of the Saffman–Taylor formulation and of the single-strip ansatz of [LM26]. The new results are the following.

- The opening range is enlarged from $0 < \kappa < -\sigma_0/(2(1-A\rho_{\mathrm{m}}))$, i.e. $c < -\frac{1}{2}\sigma_0$ in the notation of [LM26] with $\rho_{\mathrm{m}} = A$, to the sharp range $0 < \kappa < \kappa_{\max}$ (Theorems 1.1 and 1.2).
- Data need not be fully unstable: it suffices that $\Pi(z_0) = \emptyset$ (Corollary 1.3), which quantifies the pinch-singularity heuristics of [Men22] for curved interfaces.
- For $A = 0$ this gives mixing zones along the *whole* of an overturned interface (Corollary 1.4); compare with the localised mixing zones of [CFM22].
- Injection-driven viscous fingering without density contrast is covered (Corollary 1.5).
- The intermediate density ρ_{m} is arbitrary, which produces a one-parameter family of asymmetric mixing zones; [LM26] take $\rho_{\mathrm{m}} = A$.

Organisation. Section 2 recalls periodic Cauchy integrals and the Saffman–Taylor formulation, and relates the vortex-sheet strength to σ. Section 3 contains the relaxation and the h-principle. Section 4 introduces the single-strip ansatz. Section 5 proves Theorem 1.1, Section 6 proves Theorem 1.2, and Section 7 proves the corollaries and discusses an example. Section 8 contains remarks and open problems, and the appendix collects facts on Cauchy integrals.

2. Cauchy integrals and the Saffman–Taylor formulation of the Muskat problem

2.1. **Curves and notation.** We identify $\mathbb{R}^2$ with $\mathbb{C}$, $x = x_1 + ix_2$. For a vector $w = (w_1, w_2)$ we write $w^\flat := w_1 - iw_2$ (the complex conjugate of w seen as a complex number); a vector field w is divergence-free and curl-free in an open set if and only if $w^\flat$ is holomorphic there. We write $w^\perp = (-w_2, w_1)$.

Definition 2.1. For $k \ge 1$ and $\gamma \in (0,1)$ let $\mathcal{Z}^{k,\gamma}$ be the set of maps $z \in C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R};\mathbb{R}^2)$ such that $z(\alpha) - (\alpha, 0)$ is 2π-periodic and

$$\delta(z) := \inf\left\{ \frac{|\sin\frac{z(\alpha)-z(\beta)}{2}|}{|\sin\frac{\alpha-\beta}{2}|} : \ \alpha,\beta \in \mathbb{R},\ \alpha-\beta \notin 2\pi\mathbb{Z} \right\} > 0,$$

where sin is the complex sine. We norm $\mathcal{Z}^{k,\gamma}$ by $\|z - (\cdot, 0)\|_{C^{k,\gamma}(\mathbb{T})}$.

Since $|\sin\frac{w}{2}|^2 = \sin^2\frac{\operatorname{Re} w}{2} + \sinh^2\frac{\operatorname{Im} w}{2}$, the condition $\delta(z) > 0$ says that z induces an embedding of $\mathbb{T}$ into the cylinder $\mathbb{T} \times \mathbb{R}$, quantitatively; in particular $|\partial_\alpha z| \ge \delta(z)$. The curve $\Gamma = z(\mathbb{R})$ is invariant under $x \mapsto x + 2\pi e_1$ and $\mathbb{R}^2 \setminus \Gamma$ has two connected components $\Omega^+(z)$ and $\Omega^-(z)$, containing $\{x_2 > \max z_2\}$ and $\{x_2 < \min z_2\}$ respectively. We write

$$\tau = \frac{\partial_\alpha z}{|\partial_\alpha z|}, \qquad n = \tau^\perp \ \text{ (so } n = i\tau \text{ as complex numbers)}, \qquad \mathrm{d}s = |\partial_\alpha z|\,\mathrm{d}\alpha,$$

so that n points into $\Omega^+(z)$, and $n_2 = \partial_\alpha z_1/|\partial_\alpha z|$. For a function w defined near Γ we denote by $w^\pm(\alpha)$ its limits at $z(\alpha)$ from $\Omega^\pm(z)$, by $[\![w]\!] = w^+ - w^-$ its jump and by $\langle w\rangle = \frac{1}{2}(w^+ + w^-)$ its average. The constant C may change from line to line.

2.2. **Periodic Cauchy integrals.**

Definition 2.2. Let $z \in \mathcal{Z}^{1,\gamma}$ and $f \in C^{0,\gamma}(\mathbb{T};\mathbb{R}^2)$. The *Cauchy field* of f along z is the vector field $\mathcal{C}[z;f]$ on $\mathbb{R}^2 \setminus \Gamma$ defined by

$$\mathcal{C}[z;f]^\flat(x) = \frac{1}{4\pi i}\int_{\mathbb{T}} f^\flat(\beta)\,\partial_\beta z(\beta)\,\cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta. \tag{2.1}$$

Since $\frac{1}{2}\cot\frac{w}{2} = \frac{1}{w} + O(w)$, formula (2.1) is the periodisation of the classical Cauchy integral

$$\frac{1}{2\pi i}\int \frac{f^\flat(\zeta)\,\mathrm{d}\zeta}{\zeta - x}.$$

Lemma 2.3. *Let $z \in \mathcal{Z}^{1,\gamma}$, $f \in C^{0,\gamma}(\mathbb{T};\mathbb{R}^2)$ and $\mathcal{C} = \mathcal{C}[z;f]$.*

(a) *$\mathcal{C}$ is real analytic and 2π-periodic in x_1 on $\mathbb{R}^2\setminus\Gamma$, with $\operatorname{div}\mathcal{C} = \operatorname{curl}\mathcal{C} = 0$ there.*

(b) (Plemelj–Privalov) *The restrictions of $\mathcal{C}$ to $\Omega^\pm(z)$ extend to $C^{0,\gamma}(\overline{\Omega^\pm(z)})$ with*

$$\|\mathcal{C}\|_{C^{0,\gamma}(\overline{\Omega^\pm})} \le C\|f\|_{C^{0,\gamma}},$$

where C depends on γ, $\delta(z)$ and $\|z\|_{\mathcal{Z}^{1,\gamma}}$. Moreover $[\![\mathcal{C}]\!] = f$, and $\langle\mathcal{C}\rangle$ is given by (2.1) with $x = z(\alpha)$ and the integral understood in the principal value sense.

(c) *As $x_2 \to \pm\infty$, $\mathcal{C}(x) \to \mathcal{C}_{\pm\infty}$ uniformly in x_1, where $\mathcal{C}^\flat_{\pm\infty} = \pm\frac{1}{4\pi}\int_{\mathbb{T}} f^\flat\partial_\beta z\,\mathrm{d}\beta$. In particular, for a constant vector c we have $\mathcal{C}[z;c] = \pm\frac{c}{2}$ in $\Omega^\pm(z)$, and for a scalar ϕ*

$$\mathcal{C}[z;\phi n]_{\pm\infty} = \pm\frac{1}{4\pi}\Big(\int_{\mathbb{T}}\phi\,|\partial_\beta z|\,\mathrm{d}\beta\Big)e_2. \tag{2.2}$$

(d) *If $j \ge 0$, $z \in \mathcal{Z}^{j+1,\gamma}$ and $f \in C^{j,\gamma}$, then $\mathcal{C} \in C^{j,\gamma}(\overline{\Omega^\pm(z)})$ and $\mathcal{C}^\pm \in C^{j,\gamma}(\mathbb{T})$, with bounds.*

(e) (Representation) *Let w be a bounded vector field on $\mathbb{R}^2\setminus\Gamma$, 2π-periodic in x_1, divergence-free and curl-free in $\mathbb{R}^2\setminus\Gamma$, whose restrictions to $\Omega^\pm(z)$ extend continuously to the closures with $[\![w]\!] \in C^{0,\gamma}(\mathbb{T})$, and with $w(x)\to 0$ as $x_2 \to \pm\infty$. Then $w = \mathcal{C}[z;[\![w]\!]]$.*

Proof. (a) is immediate. (b) and (d) are the periodic versions of the Plemelj–Privalov theorem and are discussed in Appendix A. (c) When $x_2 \to +\infty$ (resp. $-\infty$), $\cot\frac{z(\beta)-x}{2} \to i$ (resp. $-i$) uniformly in β, which gives $\mathcal{C}_{\pm\infty}$. If $f = c$ is constant, then $\partial_x\int_{\mathbb{T}} c^\flat\partial_\beta z\cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta = -c^\flat\int_{\mathbb{T}}\partial_\beta\big[\cot\frac{z(\beta)-x}{2}\big]\,\mathrm{d}\beta = 0$ by periodicity, so $\mathcal{C}[z;c]$ is constant in each component and equals its limit $\pm\frac{1}{4\pi}c^\flat\cdot 2\pi = \pm\frac{c^\flat}{2}$; recall $\int_{\mathbb{T}}\partial_\beta z\,\mathrm{d}\beta = 2\pi$. For $f = \phi n$ we have $n^\flat\partial_\beta z = -i|\partial_\beta z|$, so $\mathcal{C}^\flat_{+\infty} = -\frac{i}{4\pi}\int\phi|\partial_\beta z|$, i.e. (2.2). (e) The field $W = w - \mathcal{C}[z;[\![w]\!]]$ is bounded, periodic, continuous across Γ, and $W^\flat$ is holomorphic off Γ; hence $W^\flat$ is entire (Morera), and by Liouville's theorem on the cylinder it is constant. By (c), $W \to -\mathcal{C}_{+\infty}$ as $x_2\to+\infty$ and $W \to -\mathcal{C}_{-\infty} = \mathcal{C}_{+\infty}$ as $x_2\to-\infty$. Therefore $\mathcal{C}_{+\infty} = 0$ and $W \equiv 0$. □

2.3. **The adjoint Neumann–Poincaré operator.**

Definition 2.4. For $z \in \mathcal{Z}^{2,\gamma}$ and a real function $\phi \in C^{0,\gamma}(\mathbb{T})$ we set

$$\mathcal{K}(z)\phi := \langle\mathcal{C}[z;\phi n]\rangle\cdot n.$$

Since the jump of $\mathcal{C}[z;\phi n]$ is normal, Lemma 2.3(b) gives

$$\mathcal{C}[z;\phi n]^\pm\cdot n = \mathcal{K}(z)\phi \pm \tfrac{1}{2}\phi, \qquad \mathcal{C}[z;\phi n]^+\cdot\tau = \mathcal{C}[z;\phi n]^-\cdot\tau. \tag{2.3}$$

In potential-theoretic terms, $\mathcal{C}[z;\phi n]$ is the gradient of the single layer potential of ϕ and $\mathcal{K}(z)$ is the (periodic) adjoint Neumann–Poincaré operator.

Lemma 2.5. *Let $k \ge 2$ and $z \in \mathcal{Z}^{k,\gamma}$.*

(a) $\mathcal{K}(z)\phi(\alpha) = \int_{\mathbb{T}} k_z(\alpha,\beta)\phi(\beta)\,\mathrm{d}\beta$, *with*

$$k_z(\alpha,\beta) = \frac{|\partial_\beta z(\beta)|}{4\pi|\partial_\alpha z(\alpha)|}\ \mathrm{Im}\left[\partial_\alpha z(\alpha)\cot\frac{z(\beta)-z(\alpha)}{2}\right],$$

and k_z extends from $\{\alpha \neq \beta\}$ to a function in $C^{k-2,\gamma}(\mathbb{T}\times\mathbb{T})$. Consequently $\mathcal{K}(z)$ maps $C^0(\mathbb{T})$ boundedly into $C^{k-2,\gamma}(\mathbb{T})$, and for $k \geq 3$ it is compact on $C^{j,\gamma}(\mathbb{T})$ for $0 \leq j \leq k-3$.

(b) $\int_{\mathbb{T}} \big(\mathcal{K}(z)\phi\big)\,|\partial_\alpha z|\,\mathrm{d}\alpha = 0$ *for every* $\phi \in C^{0,\gamma}(\mathbb{T})$.

Proof. (a) The complex density of ϕn is $\phi\, n^\flat \partial_\beta z = -i\phi|\partial_\beta z|$. Using $w\cdot n = \mathrm{Re}(w^\flat n)$ and $n = i\partial_\alpha z/|\partial_\alpha z|$, the principal value formula of Lemma 2.3(b) gives the stated kernel, once we show it is bounded (then no principal value is needed). Away from the diagonal $\{\alpha-\beta \in 2\pi\mathbb{Z}\}$ the kernel is $C^{k-1,\gamma}$, because $\delta(z) > 0$. Near the diagonal write $\cot\frac{w}{2} = \frac{2}{w} + g(w)$ with g analytic for $|w| < 2\pi$, and

$$z(\beta)-z(\alpha) = (\beta-\alpha)Z_1(\alpha,\beta), \qquad \partial_\alpha z(\alpha) - Z_1(\alpha,\beta) = -(\beta-\alpha)Z_2(\alpha,\beta),$$

with $Z_1 = \int_0^1 \partial_\alpha z(\alpha + s(\beta-\alpha))\,\mathrm{d}s \in C^{k-1,\gamma}$, $Z_2 = \int_0^1\!\int_0^1 s\,\partial_\alpha^2 z(\alpha + rs(\beta-\alpha))\,\mathrm{d}r\,\mathrm{d}s \in C^{k-2,\gamma}$ and $|Z_1| \geq c > 0$. Then

$$\mathrm{Im}\left[\frac{2\partial_\alpha z(\alpha)}{z(\beta)-z(\alpha)}\right] = \frac{2}{\beta-\alpha}\,\mathrm{Im}\left[\frac{\partial_\alpha z(\alpha)-Z_1}{Z_1}\right] = -2\,\mathrm{Im}\left[\frac{Z_2}{Z_1}\right] \in C^{k-2,\gamma},$$

while $\partial_\alpha z(\alpha) g(z(\beta)-z(\alpha)) \in C^{k-1,\gamma}$. Differentiating under the integral sign gives $\mathcal{K}(z) : C^0 \to C^{k-2,\gamma}$, and compactness follows from the compact embedding $C^{k-2,\gamma}(\mathbb{T}) \subset C^{j,\gamma}(\mathbb{T})$ for $j \leq k-3$.

(b) Let $w = \mathcal{C}[z;\phi n]$. Since w is divergence-free and periodic, its flux through Γ from above equals its flux through $\{x_2 = L\}$ for every large L, and letting $L\to\infty$ we get from (2.2) $\int_{\mathbb{T}} w^+\cdot n\,\mathrm{d}s = 2\pi(\mathcal{C}_{+\infty})_2 = \frac{1}{2}\int\phi\,\mathrm{d}s$. Similarly $\int_{\mathbb{T}} w^-\cdot n\,\mathrm{d}s = 2\pi(\mathcal{C}_{-\infty})_2 = -\frac{1}{2}\int\phi\,\mathrm{d}s$. Adding, $\int\langle w\rangle\cdot n\,\mathrm{d}s = 0$. □

Proposition 2.6. *Let $|A| < 1$, $k \geq 3$ and $z \in \mathcal{Z}^{k,\gamma}$. Then $I - 2A\mathcal{K}(z)$ is an isomorphism of $C^{j,\gamma}(\mathbb{T})$ for $0 \leq j \leq k-3$. Moreover, if $f \in C^{j,\gamma}(\mathbb{T})$ with $0 \leq j \leq k-2$, the solution $\sigma \in C^{0,\gamma}(\mathbb{T})$ of $(I-2A\mathcal{K}(z))\sigma = f$ belongs to $C^{j,\gamma}(\mathbb{T})$, with $\|\sigma\|_{C^{j,\gamma}} \leq C\|f\|_{C^{j,\gamma}}$, and $\int\sigma\,\mathrm{d}s = \int f\,\mathrm{d}s$.*

Proof. By Lemma 2.5(a) and the Riesz–Schauder theory, $I-2A\mathcal{K}(z)$ is a Fredholm operator of index zero on $C^{j,\gamma}(\mathbb{T})$, $0 \leq j \leq k-3$; so it suffices to prove injectivity on $C^{0,\gamma}$. Let $\phi \in C^{0,\gamma}$ with $\phi = 2A\mathcal{K}(z)\phi$. If $A = 0$ there is nothing to prove, so let $A \neq 0$. By Lemma 2.5(b), $\int\phi\,\mathrm{d}s = 0$, so $w := \mathcal{C}[z;\phi n]$ tends to 0 as $x_2 \to \pm\infty$, by (2.2); being the conjugate of a periodic holomorphic function, it does so exponentially fast. Since the jump of w is normal, w is curl-free in $\mathcal{D}'(\mathbb{R}^2)$, and since its circulation along horizontal lines tends to zero, $w = \nabla\Phi$ for a periodic, Lipschitz function Φ which is $C^{1,\gamma}$ up to Γ from both sides and bounded. By (2.3) and $\mathcal{K}(z)\phi = \phi/(2A)$,

$$\frac{w^+\cdot n}{1+A} = \frac{w^-\cdot n}{1-A} = \frac{\phi}{2A} =: g.$$

Integrating by parts on one period of $\Omega^\pm \cap \{|x_2| < L\}$ and letting $L\to\infty$ (the contributions of the horizontal boundaries vanish by the exponential decay, the lateral ones cancel by periodicity),

$$\frac{1}{1+A}\int_{\Omega^+}|w|^2 + \frac{1}{1-A}\int_{\Omega^-}|w|^2 = -\int_{\mathbb{T}}\Phi\,\frac{w^+\cdot n}{1+A}\,\mathrm{d}s + \int_{\mathbb{T}}\Phi\,\frac{w^-\cdot n}{1-A}\,\mathrm{d}s = 0,$$

where the integrals over $\Omega^\pm$ are over one period. Hence $w \equiv 0$ and $\phi = [\![w]\!]\cdot n = 0$.

For the regularity statement with $j = k-2$: the solution $\sigma \in C^{k-3,\gamma}$ satisfies $\sigma = f + 2A\mathcal{K}(z)\sigma$, and $\mathcal{K}(z)\sigma \in C^{k-2,\gamma}$ by Lemma 2.5(a). Finally $\int\sigma\,\mathrm{d}s = \int f\,\mathrm{d}s$ by Lemma 2.5(b). □

2.4. **The Muskat problem with viscosity jump.** Let $z \in \mathcal{Z}^{k,\gamma}$, $k \geq 3$, and $\rho = \pm 1$ in $\Omega^{\pm}(z)$. The *instantaneous Darcy problem* consists in finding u and p, $C^{0,\gamma}$ and $C^{1,\gamma}$ up to Γ from both sides, such that

$$\operatorname{div} u = 0, \quad (1 \pm A)\, u = -\nabla p \mp e_2 \ \text{ in } \Omega^{\pm}(z), \qquad [\![p]\!] = 0, \ \ [\![u \cdot n]\!] = 0 \ \text{ on } \Gamma, \tag{2.4}$$

with $u \to 0$ and $\nabla p + \rho e_2 \to 0$ as $x_2 \to \pm\infty$ (the fluids are at rest at infinity). We introduce the *Darcy field* and the *Saffman–Taylor function*

$$v := \mu(\rho) u = -\nabla p - \rho e_2, \qquad \sigma := [\![\partial_n p]\!] = \partial_n p^+ - \partial_n p^- \ \text{ on } \Gamma.$$

Proposition 2.7. *Let $|A| < 1$, $k \geq 3$, $z \in \mathcal{Z}^{k,\gamma}$, and let $\sigma \in C^{k-2,\gamma}(\mathbb{T})$ be the solution of*

$$\big(I - 2A\mathcal{K}(z)\big)\sigma = -2n_2. \tag{2.5}$$

Then (2.4) has a solution, unique up to an additive constant in p, given by

$$v = -\mathcal{C}[z; \sigma n] \mp e_2 \quad \text{and} \quad u = \frac{v}{1 \pm A} \qquad \text{in } \Omega^{\pm}(z). \tag{2.6}$$

Its Saffman–Taylor function is σ, the normal velocity $U := u \cdot n$ on Γ is

$$U = -\mathcal{K}(z)\sigma, \tag{2.7}$$

and moreover $\sigma = -2(n_2 + AU)$ and $\int_{\mathbb{T}} \sigma \, \mathrm{d}s = -4\pi$.

Proof. Define v and u by (2.6). By (2.3), $v^{\pm} \cdot n = -\mathcal{K}\sigma \mp \frac{1}{2}\sigma \mp n_2$, and (2.5) says $\frac{1}{2}\sigma + n_2 = A\mathcal{K}\sigma$. Therefore $v^{\pm} \cdot n = -(1 \pm A)\mathcal{K}\sigma$, i.e. $u^+ \cdot n = u^- \cdot n = -\mathcal{K}\sigma$, which gives $[\![u \cdot n]\!] = 0$ and (2.7); then $\sigma = -2n_2 + 2A\mathcal{K}\sigma = -2(n_2 + AU)$. Integrating (2.5) and using Lemma 2.5(b) and $\int n_2 \, \mathrm{d}s = \int \partial_\alpha z_1 \, \mathrm{d}\alpha = 2\pi$, we get $\int \sigma \, \mathrm{d}s = -4\pi$; by (2.2), $v \to \mp\frac{1}{4\pi}(\int \sigma \, \mathrm{d}s) e_2 \mp e_2 = 0$ as $x_2 \to \pm\infty$. Next, $v + \rho e_2 = -\mathcal{C}[z; \sigma n]$ in $\mathbb{R}^2 \setminus \Gamma$; this field is curl-free in $\mathcal{D}'(\mathbb{R}^2)$, because its jump is normal, and its circulation along horizontal lines vanishes (its limits at $\pm\infty$ are vertical). Hence $v + \rho e_2 = -\nabla p$ for a continuous function p, periodic in x_1, and $\partial_n p^{\pm} = \mathcal{C}[z; \sigma n]^{\pm} \cdot n = \mathcal{K}\sigma \pm \frac{1}{2}\sigma$, so that $[\![\partial_n p]\!] = \sigma$. The field u is divergence-free in each $\Omega^{\pm}$ and has no normal jump, so $\operatorname{div} u = 0$ in $\mathcal{D}'(\mathbb{R}^2)$. Regularity follows from Lemma 2.3(d).

For uniqueness, let (u', p') be the difference of two solutions. Then $\operatorname{div} u' = 0$, $(1 \pm A)u' = -\nabla p'$ in $\Omega^{\pm}$, p' is continuous, $u' \cdot n$ has no jump, and $u' \to 0$ at infinity, exponentially fast since u' is the gradient of a periodic harmonic function in $\{|x_2| > L\}$. Integrating by parts over one period,

$$\int (1 + A\rho)|u'|^2 = -\int \nabla p' \cdot u' = \int_{\mathbb{T}} p' \left(u'^{+} - u'^{-}\right) \cdot n \, \mathrm{d}s = 0,$$

so $u' = 0$ and $\nabla p' = 0$. □

Since the interface moves with the normal velocity of the fluid, Proposition 2.7 gives the following formulation of the Muskat problem with viscosity jump, in which the only operator to be inverted acts on the Saffman–Taylor function.

Corollary 2.8. *A family of curves $z(\cdot, t) \in \mathcal{Z}^{k,\gamma}$, $k \geq 3$, solves the Muskat problem with viscosity jump if and only if*

$$\partial_t z \cdot n = -\mathcal{K}(z)\sigma, \qquad \big(I - 2A\mathcal{K}(z)\big)\sigma = -2n_2.$$

Remark 2.9 (Duality with the vorticity formulation). For a real function γ, the complex densities of $\gamma\tau$ and γn are $\gamma|\partial_\beta z|$ and $-i\gamma|\partial_\beta z|$, so $\mathcal{C}[z; \gamma\tau]^\flat = i\,\mathcal{C}[z; \gamma n]^\flat$, i.e. $\mathcal{C}[z; \gamma\tau]$ is obtained by rotating $\mathcal{C}[z; \gamma n]$ by $-\pi/2$. In particular $\langle \mathcal{C}[z; \gamma\tau] \rangle \cdot \tau = \langle \mathcal{C}[z; \gamma n] \rangle \cdot n = \mathcal{K}(z)\gamma$. The velocity is curl-free off Γ with jump $\gamma\tau$, $\gamma = [\![u \cdot \tau]\!]$, hence $u = \mathcal{C}[z; \gamma\tau]$ by Lemma 2.3(e): $\langle u \rangle$ is the Birkhoff–Rott integral of the vortex sheet. The tangential part of Darcy's law, $[\![v \cdot \tau]\!] = -[\![\rho]\!]\tau_2 = -2\tau_2$, reads $\gamma + 2A\langle u \rangle \cdot \tau = -2\tau_2$, i.e. $(I + 2A\mathcal{K}(z))\gamma = -2\tau_2$; with $\varpi = \gamma|\partial_\alpha z|$ this is the familiar equation $\varpi + 2A\,\mathrm{BR}(z, \varpi) \cdot \partial_\alpha z = -2\partial_\alpha z_2$ of [CCG11]. Thus the vorticity is obtained by inverting $I + 2A\mathcal{K}(z)$, the Saffman–Taylor function by inverting $I - 2A\mathcal{K}(z)$; both are covered by Proposition 2.6. Note however that in the vorticity formulation the normal velocity $U = \langle u \rangle \cdot n = -\langle \mathcal{C}[z; \gamma n] \rangle \cdot \tau$ is a singular integral of γ, whereas in (2.7) it is the smoothing

operator $\mathcal{K}(z)$ applied to σ. More importantly, the Saffman–Taylor formulation extends to subsolutions, where the vorticity formulation does not (Section 4).

2.5. **The vortex-sheet strength and the pinch set.** For $z \in \mathcal{Z}^{2,\gamma}$ and a real function $\phi \in C^{0,\gamma}(\mathbb{T})$ we also set

$$\mathcal{T}(z)\phi := \langle \mathcal{C}[z;\phi n]\rangle \cdot \tau.$$

Unlike $\mathcal{K}(z)$, the operator $\mathcal{T}(z)$ is a singular integral operator (a Hilbert transform plus a smoothing operator); it is bounded on $C^{j,\gamma}(\mathbb{T})$ for $z \in \mathcal{Z}^{j+1,\gamma}$ by Lemma 2.3(d). The vortex-sheet strength of the Muskat problem is a local function of σ and $\mathcal{T}(z)\sigma$:

Proposition 2.10. *In the setting of Proposition 2.7, the vortex-sheet strength $\gamma = [\![u\cdot\tau]\!]$ is*

$$\gamma = -\frac{2\big(\tau_2 - A\,\mathcal{T}(z)\sigma\big)}{1-A^2}, \tag{2.8}$$

and it is the unique solution in $C^{0,\gamma}(\mathbb{T})$ of $(I+2A\mathcal{K}(z))\gamma = -2\tau_2$. In particular $\gamma \in C^{k-2,\gamma}(\mathbb{T})$.

Proof. By (2.6), $v^{\pm}\cdot\tau = -\mathcal{T}(z)\sigma \mp \tau_2$ (the tangential traces of $\mathcal{C}[z;\sigma n]$ coincide). Since $u^{\pm} = v^{\pm}/(1\pm A)$,

$$\gamma = \frac{-\mathcal{T}\sigma-\tau_2}{1+A} - \frac{-\mathcal{T}\sigma+\tau_2}{1-A} = \frac{2A\,\mathcal{T}\sigma - 2\tau_2}{1-A^2}.$$

The velocity is curl-free and divergence-free off Γ, has no normal jump, tends to 0 at infinity and has tangential jump γ, so $u = \mathcal{C}[z;\gamma\tau]$ by Lemma 2.3(e); Remark 2.9 then gives $(I+2A\mathcal{K}(z))\gamma = -2\tau_2$, and uniqueness and regularity follow from Proposition 2.6 applied with $-A$ in place of A. □

Definition 2.11. Let $z_0 \in \mathcal{Z}^{k,\gamma}$, $k \ge 3$, and $\mathcal{K}_0 := \mathcal{K}(z_0)$, $\mathcal{T}_0 := \mathcal{T}(z_0)$. The Saffman–Taylor function, the normal velocity and the vortex-sheet strength of the Muskat data $\rho_0 = \mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}$ are

$$\sigma_0 := -2\big(I-2A\mathcal{K}_0\big)^{-1}[n_{0,2}], \qquad U_0 := -\mathcal{K}_0\sigma_0, \qquad \gamma_0 := -2\big(I+2A\mathcal{K}_0\big)^{-1}[\tau_{0,2}].$$

The data are *fully unstable* if $\sigma_0 < 0$ on $\mathbb{T}$. The *pinch set* is $\Pi(z_0) := \{\alpha :\ \sigma_0(\alpha)\ge 0,\ \gamma_0(\alpha) = 0\}$.

Example 2.12. For the flat interface $z_0(\alpha) = (\alpha, 0)$ the kernel k_{z_0} vanishes (the cotangent of a real number is real), so $\mathcal{K}_0 = 0$, $\sigma_0 \equiv -2$, $U_0 \equiv 0$ and $\gamma_0 \equiv 0$. If $A = 0$, then $\sigma_0 = -2n_{0,2}$ and $\gamma_0 = -2\tau_{0,2}$ for every z_0: the data are fully unstable if and only if z_0 is a graph, and $\Pi(z_0) = \{n_0 = -e_2\} = \{\tau_0 = -e_1\}$.

For $z_0 \in \mathcal{Z}^{6,\gamma}$, Propositions 2.6 (with $k=6$, $j=4$) and 2.10, and Lemma 2.5(a), give $\sigma_0, U_0, \gamma_0 \in C^{4,\gamma}(\mathbb{T})$. The quantities Φ and $\kappa_{\max}$ of (1.5) are then well defined, $\kappa_{\max} = 0$ exactly on $\Pi(z_0)$, and since the expression under the square root can only vanish where $\sigma_0 = \gamma_0 = 0$, we have $\kappa_{\max} \in C^{4,\gamma}(\mathbb{T})$ whenever $\Pi(z_0) = \emptyset$.

3. Relaxation of IPM with viscosity jump

3.1. **IPM as a differential inclusion.** A pair $(\rho, u) \in L^\infty(\mathbb{T}\times\mathbb{R}\times(0,T))^{1+2}$ with $\rho^2 = 1$ a.e. is a *weak solution* of (1.1) with initial datum ρ_0 if for every $\varphi \in C_c^\infty(\mathbb{T}\times\mathbb{R}\times[0,T))$

$$\begin{aligned} &\int_0^T\!\!\int \big(\rho\,\partial_t\varphi + \rho u\cdot\nabla\varphi\big)\,\mathrm{d}x\,\mathrm{d}t + \int \rho_0\varphi(\cdot,0)\,\mathrm{d}x = 0, \qquad \int_0^T\!\!\int u\cdot\nabla\varphi\,\mathrm{d}x\,\mathrm{d}t = 0,\\ &\int_0^T\!\!\int \big((1+A\rho)\,u\cdot\nabla^\perp\varphi + \rho\,\partial_1\varphi\big)\,\mathrm{d}x\,\mathrm{d}t = 0, \end{aligned} \tag{3.1}$$

where $\nabla^\perp = (-\partial_2, \partial_1)$; the last identity is the curl of Darcy's law, $\operatorname{curl}(\mu(\rho)u) = -\partial_1\rho$. Consider the linear system for $(\rho, u, m) \in \mathbb{R}\times\mathbb{R}^2\times\mathbb{R}^2$

$$\partial_t\rho + \operatorname{div} m = 0, \qquad \operatorname{div} u = 0, \qquad \operatorname{curl}(u + Am) = -\partial_1\rho, \tag{3.2}$$

and the constitutive set

$$\mathsf{K} := \{(\rho, u, m) : \ \rho \in \{-1, 1\}, \ m = \rho u\}.$$

Since $(1 + A\rho)u = u + Am$ on K, a pair (ρ, u) is a weak solution if and only if $(\rho, u, \rho u)$ solves (3.2) in the sense of distributions, with initial datum ρ_0, and takes values in K a.e. This linearisation of Darcy's law is due to [Men22].

For a state (ρ, u, m) we define the *Darcy field* $v := u + Am$. In terms of (ρ, v, m) the system (3.2) becomes

$$\partial_t \rho + \operatorname{div} m = 0, \qquad \operatorname{curl} v = -\partial_1 \rho, \qquad \operatorname{div} v = -A\,\partial_t \rho, \qquad u = v - Am. \tag{3.3}$$

The point of (3.3) is that v *is determined by* ρ *and* $\partial_t \rho$ *alone* (modulo harmonic fields), whereas the vorticity of $u = v - Am$ depends on $\operatorname{curl} m$. On K we have $v = (1 + A\rho)u = \mu(\rho)u = -\nabla p - \rho e_2$.

3.2. **Laminates.** The wave cone of (3.2) is the set Λ of states $\bar{W} = (\bar\rho, \bar u, \bar m)$ for which there exist $\xi \in \mathbb{R}^2 \setminus \{0\}$ and $c \in \mathbb{R}$ such that $\bar W h(\xi \cdot x + ct)$ solves (3.2) for every $h \in C^1(\mathbb{R})$, that is,

$$c\bar\rho + \xi \cdot \bar m = 0, \qquad \xi \cdot \bar u = 0, \qquad \xi^\perp \cdot (\bar u + A\bar m) + \xi_1 \bar\rho = 0. \tag{3.4}$$

For a state (ρ, u, m) with $|\rho| < 1$ we introduce

(3.5)

$$\mu := 1 + A\rho, \quad v := u + Am, \quad d := \frac{2\big[(1 - A^2)m - (\rho - A)v\big]}{1 - \rho^2}, \quad D := d + 2e_2, \quad a := e_2 + \frac{A}{\mu} v.$$

For fixed ρ the map $(u, m) \mapsto (v, d)$ is a linear isomorphism of $\mathbb{R}^4$, with inverse

$$m = \frac{(\rho - A)v + \frac12(1 - \rho^2)d}{1 - A^2}, \qquad u = v - Am.$$

Lemma 3.1 (First order laminates)**.** *Let $\rho \in (-1, 1)$, $\lambda_\pm = \frac{1 \pm \rho}{2}$, and $P_+ = (1, u_+, u_+)$, $P_- = (-1, u_-, -u_-) \in \mathsf{K}$. Then $P_+ - P_- \in \Lambda$ if and only if there is a unit vector ν with*

$$u_+ \cdot \nu = u_- \cdot \nu, \qquad \big((1 + A)u_+ - (1 - A)u_-\big) \cdot \nu^\perp = -2\nu_1. \tag{3.6}$$

In that case the barycentre $P = \lambda_+ P_+ + \lambda_- P_- = (\rho, u, m)$ satisfies

$$D = 2(a \cdot \nu)\nu, \qquad \textit{in particular} \qquad |D - a| = |a|. \tag{3.7}$$

Conversely, for every $\rho \in (-1, 1)$, $v \in \mathbb{R}^2$ and unit vector ν, the state with these values of ρ, v and with $D = 2(a \cdot \nu)\nu$ is such a barycentre.

Proof. We have $P_+ - P_- = (2, u_+ - u_-, u_+ + u_-)$. Taking $\xi = \nu$ in (3.4), the first condition only fixes c, the second is $\nu \cdot (u_+ - u_-) = 0$ and the third is $\nu^\perp \cdot (u_+ - u_- + A(u_+ + u_-)) + 2\nu_1 = 0$, which is (3.6). Let $v_\pm := (1 \pm A)u_\pm$ be the Darcy fields of the two phases and $\delta := v_+ - v_-$. Then $v = u + Am = \lambda_+ v_+ + \lambda_- v_-$, so $v_+ = v + \lambda_- \delta$, $v_- = v - \lambda_+ \delta$, and

$$(1 - A^2)m = (1 - A)\lambda_+ v_+ - (1 + A)\lambda_- v_- = (\rho - A)v + 2\lambda_+\lambda_-\delta = (\rho - A)v + \tfrac12(1 - \rho^2)\delta,$$

that is, $d = \delta$. The first condition in (3.6) reads $(1 - A)v_+ \cdot \nu = (1 + A)v_- \cdot \nu$, i.e. $\delta \cdot \nu\,[(1 - A)\lambda_- + (1 + A)\lambda_+] = 2A\, v \cdot \nu$; since $(1 - A)\lambda_- + (1 + A)\lambda_+ = \mu$, this is $\delta \cdot \nu = \frac{2A}{\mu} v \cdot \nu$. The second condition reads $\delta \cdot \nu^\perp = -2\nu_1 = -2e_2 \cdot \nu^\perp$. Hence

$$\delta = \frac{2A}{\mu}(v \cdot \nu)\nu - 2(e_2 \cdot \nu^\perp)\nu^\perp = -2e_2 + 2\Big(\big(e_2 + \tfrac{A}{\mu} v\big) \cdot \nu\Big)\nu,$$

where we used $\nu^\perp \otimes \nu^\perp = I - \nu \otimes \nu$. This is (3.7), and $|2(a \cdot \nu)\nu - a|^2 = |a|^2$. The computation can be reversed: given (ρ, v, ν) put $\delta = -2e_2 + 2(a \cdot \nu)\nu$, $v_+ = v + \lambda_- \delta$, $v_- = v - \lambda_+ \delta$ and $u_\pm = v_\pm/(1 \pm A)$. □

Proposition 3.2 (The relaxed set)**.** *The set*

$$\mathcal{U}_A := \big\{(\rho,u,m)\in\mathbb{R}\times\mathbb{R}^2\times\mathbb{R}^2 :\ |\rho|<1,\ |D-a|<|a|\big\} = \big\{|\rho|<1,\ D\cdot(D-2a)<0\big\},$$

with D, a given by (3.5), is open, and every point of $\mathcal{U}_A$ is the barycentre of a Λ-segment whose endpoints are first order laminates of K as in Lemma 3.1. In particular $\mathcal{U}_A$ is contained in the lamination convex hull K^{lc}, hence in the interior of the Λ-convex hull K^{Λ}.

Proof. Openness is clear. For $\bar m\neq 0$ the state $\bar W=(0,-A\bar m,\bar m)$ belongs to Λ: take $\xi=\bar m^{\perp}$ and any c in (3.4). Moving along $\bar W$ leaves ρ and $v=u+Am$ (hence a) unchanged and moves d, hence D, in the direction $\bar m$. Let $(\rho,u,m)\in\mathcal{U}_A$; then D lies in the open disc $\{|D'-a|<|a|\}$, and any line through D meets the circle $\{|D'-a|=|a|\}$ at two points D_1, D_2 with $D\in(D_1,D_2)$. The circle is exactly $\{2(a\cdot\nu)\nu :\ |\nu|=1\}$ (the circle through the origin with centre a), so by the converse part of Lemma 3.1 the states with values (ρ,v,D_j) are first order laminates. The equivalence of the two descriptions of $\mathcal{U}_A$ follows from $|D-a|^2-|a|^2=D\cdot(D-2a)$. □

Remark 3.3. (a) For $A=0$ we have $v=u$, $a=e_2$ and $D-a=\frac{2}{1-\rho^2}\big(m-\rho u+\frac12(1-\rho^2)e_2\big)$, so $\mathcal{U}_0$ is the interior of the hull $\{|m-\rho u+\frac12(1-\rho^2)e_2|\le\frac12(1-\rho^2)\}$ computed by Székelyhidi [Sze12].

(b) The disc degenerates when $a=0$, i.e. $v=-\frac{\mu}{A}e_2$. For a state in a thin mixing zone around an interface, $a\cdot n$ is (up to lower order terms) $-\frac12\sigma$ and $a\cdot\tau$ is related to the vorticity of the sheet (see Lemma 5.2); thus mixing is impossible where there is neither Saffman–Taylor instability nor vorticity, in agreement with the pinch singularity observed in [Men22].

(c) We only need $\mathcal{U}_A\subset\operatorname{int}\mathsf{K}^{\Lambda}$; we do not claim that $\mathcal{U}_A$ coincides with the interior of the Λ-convex hull.

3.3. Subsolutions and the h-principle.

Definition 3.4. Let $T>0$ and $\rho_0\in L^\infty(\mathbb{T}\times\mathbb{R})$ with $\rho_0^2=1$. A triple $(\bar\rho,\bar u,\bar m)\in L^\infty(\mathbb{T}\times\mathbb{R}\times(0,T))^{1+2+2}$ is a *subsolution* with initial datum ρ_0 and mixing zone Ω_{mix}, an open subset of $\mathbb{T}\times\mathbb{R}\times(0,T)$, if

(i) (3.2) holds in the sense of distributions on $\mathbb{T}\times\mathbb{R}\times[0,T)$ with initial datum ρ_0, i.e. for every $\varphi\in C_c^\infty(\mathbb{T}\times\mathbb{R}\times[0,T))$

$$\iint\big(\bar\rho\partial_t\varphi+\bar m\cdot\nabla\varphi\big)+\int\rho_0\varphi(\cdot,0)=0,\qquad \iint\bar u\cdot\nabla\varphi=0,\qquad \iint\big((\bar u+A\bar m)\cdot\nabla^{\perp}\varphi+\bar\rho\,\partial_1\varphi\big)=0;$$

(ii) $(\bar\rho,\bar u,\bar m)$ is continuous on Ω_{mix} and takes values in $\mathcal{U}_A$ there;

(iii) $(\bar\rho,\bar u,\bar m)\in\mathsf{K}$ a.e. in the complement of Ω_{mix}.

The following h-principle was proved by Mengual [Men22] for $|A|<1$ (for $A=0$ see [CFG11, Sze12, CCF21]); we state it in the form in which we use it. The convex integration scheme is local in space-time, so the horizontally periodic setting plays no role, and by Proposition 3.2 subsolutions in the sense of Definition 3.4 take values in the interior of the Λ-convex hull, as required in [Men22]. We refer to [Men22] for the precise functional framework.

Theorem 3.5 (h-principle [Men22])**.** *Let $(\bar\rho,\bar u,\bar m)$ be a subsolution with initial datum ρ_0 and mixing zone Ω_{mix}. Then there exist infinitely many weak solutions (ρ,u) of (1.1) with initial datum ρ_0 such that $(\rho,u)=(\bar\rho,\bar u)$ a.e. outside Ω_{mix} and ρ mixes in Ω_{mix}, i.e. both $\{\rho=1\}$ and $\{\rho=-1\}$ have positive measure in every open subset of Ω_{mix}. Moreover there is a sequence of such solutions with $(\rho_k,u_k)\overset{*}{\rightharpoonup}(\bar\rho,\bar u)$ in L^∞.*

3.4. The flat interface.

The following explicit subsolutions illustrate the role of the viscosity jump; the flat case is treated in [Men22].

Example 3.6. Let $\theta\in(0,1)$ and $F_\theta(\rho):=\theta\frac{1-\rho^2}{1-A\rho}$ for $\rho\in[-1,1]$. Then $F_\theta(\pm1)=0$,

$$F_\theta'(\rho)=\theta\,\frac{A-2\rho+A\rho^2}{(1-A\rho)^2},\qquad F_\theta''(\rho)=-\frac{2\theta(1-A^2)}{(1-A\rho)^3}<0,$$

so $-F_\theta'$ increases from $-\frac{2\theta}{1+A}$ at $\rho=-1$ to $\frac{2\theta}{1-A}$ at $\rho=1$. Let $R_\theta:\mathbb{R}\to[-1,1]$ be equal to -1 on $(-\infty,-\frac{2\theta}{1+A}]$, to 1 on $[\frac{2\theta}{1-A},\infty)$, and to the inverse of $-F_\theta'$ in between, and set

$$\bar\rho(x,t)=R_\theta(x_2/t),\qquad \bar u=0,\qquad \bar m=-F_\theta(\bar\rho)\,e_2.$$

Then $\partial_t\bar\rho+\partial_2\bar m_2=-\frac{R_\theta'}{t}\big(\frac{x_2}{t}+F_\theta'(R_\theta)\big)=0$ (a rarefaction wave), $\operatorname{div}\bar u=0$, and $\operatorname{curl}(\bar u+A\bar m)=-A\partial_1F_\theta(\bar\rho)=0=-\partial_1\bar\rho$; all fields are continuous, so (3.2) holds with initial datum $\operatorname{sign}(x_2)$. Outside the fan $\{-\frac{2\theta t}{1+A}<x_2<\frac{2\theta t}{1-A}\}$ the state is $(\pm1,0,0)\in\mathsf{K}$. Inside, $v=A\bar m$, $(1-A^2)\bar m-(\bar\rho-A)v=(1-A\bar\rho)\bar m$, hence

$$D=2(1-\theta)e_2,\qquad a=\Big(1-\frac{A^2\theta(1-\bar\rho^2)}{1-A^2\bar\rho^2}\Big)e_2,\qquad D\cdot(D-2a)=-\frac{4\theta(1-\theta)(1-A^2)}{1-A^2\bar\rho^2}<0.$$

So $(\bar\rho,\bar u,\bar m)$ is a subsolution, and Theorem 3.5 gives mixing solutions whose mixing zone is $\{-\frac{2\theta t}{1+A}<x_2<\frac{2\theta t}{1-A}\}$. For $A>0$ (the heavy fluid on top is the more viscous one) the mixing zone grows faster upwards: the less viscous light fluid fingers into the more viscous one, as predicted by Saffman and Taylor. Moreover $\int(\bar\rho-\operatorname{sign}x_2)\,\mathrm{d}x_2=t\int_{-1}^1F_\theta'(r)\,\mathrm{d}r=0$: the centre of mass of the mixing zone does not move.

4. Piecewise constant subsolutions

In this section $z_0\in\mathcal{Z}^{6,\gamma}$, $\rho_{\mathrm m}\in(-1,1)$ and $\kappa\in C^{4,\gamma}(\mathbb{T})$ with $\kappa>0$ are fixed, $\sigma_0,U_0\in C^{4,\gamma}(\mathbb{T})$ are as in Definition 2.11, and

$$\mu_{\mathrm m}:=1+A\rho_{\mathrm m},\qquad c^\pm:=(1\pm\rho_{\mathrm m})\kappa.$$

Moreover $a\in C^{3,\gamma}(\mathbb{T})$ and a polynomial h with $h(0)=h'(0)=0$ are given; they will be chosen in Section 5.

4.1. **Geometry.** We define the central curve and the two boundary curves of the mixing zone,

$$Z(\alpha,t):=z_0(\alpha)+\Big(tU_0(\alpha)+\tfrac{t^2}{2}a(\alpha)\Big)n_0(\alpha)+h(t)e_2,\qquad z^\pm(\alpha,t):=Z(\alpha,t)\pm t\,c^\pm(\alpha)\,n_0(\alpha),\tag{4.1}$$

and the map

$$X(\alpha,\lambda,t):=\tfrac12(1-\lambda)\,z^-(\alpha,t)+\tfrac12(1+\lambda)\,z^+(\alpha,t),\qquad \partial_\lambda X=\tfrac12(z^+-z^-)=t\kappa\,n_0.\tag{4.2}$$

Lemma 4.1. *There is $T_1>0$ such that for $t\in(0,T_1]$: $z^\pm(\cdot,t)\in\mathcal{Z}^{3,\gamma}$ with $\delta(z^\pm(\cdot,t))\ge\frac12\delta(z_0)$; $\Gamma^+(t)\subset\Omega^+(z^-(\cdot,t))$; and $X(\cdot,\cdot,t)$ is a $C^{3,\gamma}$ diffeomorphism from $\mathbb{T}\times[-1,1]$ onto the closure of the strip*

$$S_t:=\Omega^-(z^+(\cdot,t))\cap\Omega^+(z^-(\cdot,t)),$$

mapping $\lambda=\pm1$ onto $\Gamma^\pm(t)$. Moreover $z^\pm$ are polynomials in t with coefficients in $C^{3,\gamma}$, and $z^\pm(\cdot,t)\to z_0$ in $C^{3,\gamma}$ as $t\to0$.

Proof. All statements follow from $z^\pm(\cdot,t)-z_0=O(t)$ in $C^{3,\gamma}$, the continuity of δ with respect to the C^1 norm, $z^+-z^-=2t\kappa n_0$ with $\kappa>0$, and the fact that $\partial_\alpha X=\partial_\alpha z_0+O(t)$ and $\partial_\lambda X=t\kappa n_0$ are linearly independent; injectivity of X for small t is the usual tubular neighbourhood argument. □

We denote by $n^\pm$ the upward unit normals of $\Gamma^\pm(t)$ and by

$$V^\pm:=\partial_tz^\pm\cdot n^\pm$$

their normal velocities. Note that the central curve Z is the *centre of mass* of the strip: since $(1-\rho_{\mathrm m})c^+=(1+\rho_{\mathrm m})c^-$,

$$Z=\tfrac12\big[(1-\rho_{\mathrm m})z^++(1+\rho_{\mathrm m})z^-\big].\tag{4.3}$$

4.2. **The subsolution.** Let $[\![\rho]\!]^+ := 1-\rho_{\mathrm{m}}$ and $[\![\rho]\!]^- := 1+\rho_{\mathrm{m}}$ be the jumps of the density across $\Gamma^\pm$, and define

$$f^\pm := [\![\rho]\!]^\pm\Big(-e_2+\big(n_2^\pm+AV^\pm\big)n^\pm\Big), \qquad v := \mathcal{C}[z^+;f^+]+\mathcal{C}[z^-;f^-] \quad \text{on } \mathbb{R}^2\setminus(\Gamma^+\cup\Gamma^-). \tag{4.4}$$

Let $v_S^\pm(\alpha,t)$ be the limits of v at $z^\pm(\alpha,t)$ from inside the strip:

$$v_S^+ = \mathcal{C}[z^+;f^+]^- + \mathcal{C}[z^-;f^-]\circ z^+, \qquad v_S^- = \mathcal{C}[z^-;f^-]^+ + \mathcal{C}[z^+;f^+]\circ z^-. \tag{4.5}$$

We define the *boundary fluxes* and the *flux mismatch*

$$F^+ := |\partial_\alpha z^+|\Big\{\tfrac{2}{1+\rho_{\mathrm{m}}}\big(v_S^+\cdot n^+ - (1-A)V^+\big)+2n_2^+\Big\}, \tag{4.6}$$

$$F^- := |\partial_\alpha z^-|\Big\{\tfrac{2}{1-\rho_{\mathrm{m}}}\big((1+A)V^- - v_S^-\cdot n^-\big)+2n_2^-\Big\}, \tag{4.7}$$

$$M := F^+ - F^-, \tag{4.8}$$

the functions

$$\Psi^-(\alpha,t) := \int_0^\alpha F^-(\beta,t)\,\mathrm{d}\beta, \qquad \Psi^+(\alpha,t) := \int_0^\alpha F^+(\beta,t)\,\mathrm{d}\beta + C_\Psi(t), \tag{4.9}$$

where C_Ψ is a continuous function of time to be chosen (it fixes the relative normalisation of the two boundary values and will produce the tilt in Section 5; $C_\Psi\equiv 0$ is allowed), and the stream function Ψ on $\overline{S_t}$ by

$$\Psi\big(X(\alpha,\lambda,t),t\big) := \tfrac12(1-\lambda)\Psi^-(\alpha,t)+\tfrac12(1+\lambda)\Psi^+(\alpha,t). \tag{4.10}$$

Finally we set $\bar\rho(\cdot,0)=\rho_0$ and, for $0<t\le T_1$,

$$(\bar\rho,\bar u,\bar m) := \begin{cases} \Big(1,\ \dfrac{v}{1+A},\ \dfrac{v}{1+A}\Big) & \text{in } \Omega^+(z^+),\\[2mm] \Big(\rho_{\mathrm{m}},\ v-A\bar m,\ \dfrac{(\rho_{\mathrm{m}}-A)v+\frac12(1-\rho_{\mathrm{m}}^2)(D-2e_2)}{1-A^2}\Big), \quad D:=\nabla^\perp\Psi, & \text{in } S_t,\\[2mm] \Big(-1,\ \dfrac{v}{1-A},\ -\dfrac{v}{1-A}\Big) & \text{in } \Omega^-(z^-). \end{cases} \tag{4.11}$$

By construction, in S_t the Darcy field of the state is $\bar u + A\bar m = v$, and the quantities (3.5) are $d = D-2e_2$, $D=\nabla^\perp\Psi$ and $a = e_2+\frac{A}{\mu_{\mathrm{m}}}v$.

Lemma 4.2 (The linear equations)**.** *Assume that*

$$\int_{\mathbb{T}} M(\alpha,t)\,\mathrm{d}\alpha = 0 \qquad \textit{for all } t\in(0,T_1]. \tag{4.12}$$

Then D is well defined and 2π-periodic in x_1, $(\bar\rho,\bar u,\bar m)$ is bounded, and it solves (3.2) in the sense of distributions on $\mathbb{T}\times\mathbb{R}\times[0,T_1)$ with initial datum ρ_0. Outside the closure of $\bigcup_t S_t\times\{t\}$ it takes values in K.

Proof. Periodicity. $\Psi^\pm(\alpha+2\pi,t)=\Psi^\pm(\alpha,t)+\Phi^\pm(t)$ with $\Phi^\pm=\int_{\mathbb{T}}F^\pm$, and $\Phi^+=\Phi^-$ by (4.12). Since $X(\alpha+2\pi,\lambda,t)=X(\alpha,\lambda,t)+2\pi e_1$, we get $\Psi(x+2\pi e_1)=\Psi(x)+\Phi^+$, so $D=\nabla^\perp\Psi$ is periodic. Ψ is C^1 up to ∂S_t, hence D is bounded.

The equations in the open regions. In each of the three open regions $\bar\rho$ is constant; v is divergence-free and curl-free there by Lemma 2.3(a), and $\operatorname{div} D = \operatorname{div}\nabla^\perp\Psi = 0$. Hence $\operatorname{div}\bar m = 0$, $\operatorname{div}\bar u = \operatorname{div} v - A\operatorname{div}\bar m = 0$ and $\operatorname{curl}(\bar u + A\bar m) = \operatorname{curl} v = 0$ in each region. Outside the strip, $\bar m = \bar\rho\bar u$ with $\bar\rho=\pm1$, i.e. the state is in K.

Jump conditions. A piecewise C^1 bounded field satisfies (3.2) in $\mathcal{D}'$ if and only if it does so in each open region and satisfies the Rankine–Hugoniot conditions across the moving curves:

$$[\![\bar\rho]\!]V = [\![\bar m\cdot n]\!], \qquad [\![\bar u\cdot n]\!]=0, \qquad [\![(\bar u+A\bar m)\cdot\tau]\!] = -[\![\bar\rho]\!]\tau_2.$$

(The initial condition holds because the fields are bounded and $\bar\rho(\cdot,t)\to\rho_0$ in L^1_{loc}.) Consider Γ^+. The jump of v across Γ^+ is f^+ (the field $\mathcal{C}[z^-;f^-]$ is continuous there), and $f^+\cdot\tau^+ = -[\![\rho]\!]^+\tau_2^+$, $f^+\cdot n^+ = A[\![\rho]\!]^+V^+$. Since $\bar u+A\bar m = v$ on both sides, the third condition holds.

Next, the flux of $\nabla^\perp\Psi$ through Γ^+ per unit α is $\nabla\Psi\cdot\partial_\alpha z^+ = \partial_\alpha(\Psi\circ z^+) = \partial_\alpha\Psi^+ = F^+$, so on the strip side

$$D\cdot n^+ = \frac{F^+}{|\partial_\alpha z^+|}, \qquad (D-2e_2)\cdot n^+ = \frac{2}{1+\rho_{\mathrm{m}}}\big(v_S^+\cdot n^+ - (1-A)V^+\big),$$

and therefore

$$(1-A^2)\,\bar m^S\cdot n^+ = (\rho_{\mathrm{m}}-A)v_S^+\cdot n^+ + (1-\rho_{\mathrm{m}})\big(v_S^+\cdot n^+ - (1-A)V^+\big) = (1-A)\big(v_S^+\cdot n^+ - (1-\rho_{\mathrm{m}})V^+\big).$$

Above Γ^+, $\bar m\cdot n^+ = v^{\mathrm{above}}\cdot n^+/(1+A) = (v_S^+\cdot n^+ + A(1-\rho_{\mathrm{m}})V^+)/(1+A)$. Subtracting, $[\![\bar m\cdot n^+]\!] = (1-\rho_{\mathrm{m}})V^+ = [\![\bar\rho]\!]V^+$. Moreover $\bar u^S\cdot n^+ = v_S^+\cdot n^+ - A\bar m^S\cdot n^+ = (v_S^+\cdot n^+ + A(1-\rho_{\mathrm{m}})V^+)/(1+A) = \bar u^{\mathrm{above}}\cdot n^+$. The computation on Γ^- is identical: there $(D-2e_2)\cdot n^- = \frac{2}{1-\rho_{\mathrm{m}}}((1+A)V^- - v_S^-\cdot n^-)$, which gives $(1-A^2)\bar m^S\cdot n^- = (1+A)\big((1+\rho_{\mathrm{m}})V^- - v_S^-\cdot n^-\big)$, while below Γ^-, $\bar m\cdot n^- = -(v_S^-\cdot n^- - A(1+\rho_{\mathrm{m}})V^-)/(1-A)$; hence $[\![\bar m\cdot n^-]\!] = (1+\rho_{\mathrm{m}})V^-$ and $[\![\bar u\cdot n^-]\!] = 0$. □

Thus the ansatz (4.11) solves the linear system exactly, whatever the curves $z^\pm$ are, provided (4.12) holds; the only remaining requirement for a subsolution is the open condition $D\cdot(D-2a)<0$ in S_t. The next lemma identifies (4.12) with the conservation of mass.

Lemma 4.3 (Mass identity)**.** *Let* $\mathcal{A}^\pm(t) := \int_{\mathbb{T}} z_2^\pm\,\partial_\alpha z_1^\pm\,\mathrm{d}\alpha$, *so that* $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^\pm = \int_{\mathbb{T}} V^\pm|\partial_\alpha z^\pm|\,\mathrm{d}\alpha$, *and*

$$Q(t) := (1-\rho_{\mathrm{m}})\tfrac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^+(t) + (1+\rho_{\mathrm{m}})\tfrac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^-(t).$$

Then for $t\in(0,T_1]$

$$\int_{\mathbb{T}} M(\alpha,t)\,\mathrm{d}\alpha = -\frac{2\,Q(t)}{1-\rho_{\mathrm{m}}^2}.$$

Proof. The identity for $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^\pm$ follows from an integration by parts: $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{A} = \int(\partial_t z_2\partial_\alpha z_1 - \partial_\alpha z_2\partial_t z_1) = \int \partial_t z\cdot(\partial_\alpha z)^\perp$. By Lemma 2.3(c), the constant parts $-[\![\rho]\!]^\pm e_2$ of the densities contribute $\mp\frac12\sum_\pm[\![\rho]\!]^\pm e_2 = \mp e_2$ to $v_{\pm\infty}$, while by (2.2) the normal parts contribute $\pm\frac{1}{4\pi}\sum_\pm[\![\rho]\!]^\pm(2\pi + A\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^\pm)e_2$. Hence

$$v_{\pm\infty} = \pm\frac{AQ}{4\pi}e_2.$$

Since v is divergence-free and periodic above Γ^+, the flux of v through Γ^+ from above equals $2\pi(v_{+\infty})_2 = \frac{AQ}{2}$; subtracting the jump $f^+\cdot n^+ = A(1-\rho_{\mathrm{m}})V^+$,

$$\int_{\mathbb{T}} v_S^+\cdot n^+|\partial_\alpha z^+|\,\mathrm{d}\alpha = \frac{AQ}{2} - A(1-\rho_{\mathrm{m}})\tfrac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^+, \qquad \int_{\mathbb{T}} v_S^-\cdot n^-|\partial_\alpha z^-|\,\mathrm{d}\alpha = -\frac{AQ}{2} + A(1+\rho_{\mathrm{m}})\tfrac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^-,$$

the second identity being obtained in the same way from below. Using $\int n_2^\pm|\partial_\alpha z^\pm| = 2\pi$ we get

$$\int_{\mathbb{T}} F^+ = \frac{2}{1+\rho_{\mathrm{m}}}\Big(\frac{AQ}{2} - (1-A\rho_{\mathrm{m}})\tfrac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^+\Big) + 4\pi,$$
$$\int_{\mathbb{T}} F^- = \frac{2}{1-\rho_{\mathrm{m}}}\Big((1-A\rho_{\mathrm{m}})\tfrac{\mathrm{d}}{\mathrm{d}t}\mathcal{A}^- + \frac{AQ}{2}\Big) + 4\pi,$$

and the difference is $\frac{2}{1-\rho_{\mathrm{m}}^2}\big(-\rho_{\mathrm{m}}AQ - (1-A\rho_{\mathrm{m}})Q\big) = -\frac{2Q}{1-\rho_{\mathrm{m}}^2}$. □

Remark 4.4. $Q=0$ means that $(1-\rho_{\mathrm{m}})\mathcal{A}^+ + (1+\rho_{\mathrm{m}})\mathcal{A}^-$ is constant, i.e. that the total mass $\int(\bar\rho-\rho_{\mathrm{ref}})$ over a period is conserved. In view of (4.3), the central interface Z separates, to first order in the width of the strip, the same amounts of heavy and light fluid as the strip itself; this is why Z must move, to leading order, with the normal velocity of the Muskat problem (Lemma 5.1).

The following lemma shows that the tangential component of D is controlled by the primitive of M, and its normal component by the boundary fluxes.

Lemma 4.5 (The state in the strip)**.** *Assume* (4.12)*, and that there are* $C_1 \geq 1$*,* $F_0 \in C^0(\mathbb{T})$ *and* $\delta \in C^0(\mathbb{T})$ *such that for* $t \in (0, T_1]$

$$\begin{aligned} &\|F^{\pm}(\cdot,t) - F_0\|_{C^0(\mathbb{T})} \leq C_1 t, \qquad \|M(\cdot,t)\|_{C^0(\mathbb{T})} \leq C_1 t, \\ &\sup_{\alpha\in[0,2\pi]} \Big| \int_0^{\alpha} M(\beta,t)\,\mathrm{d}\beta + C_\Psi(t) + 2t\kappa(\alpha)\delta(\alpha)\Big| \leq C_1 t^2. \end{aligned} \tag{4.13}$$

Then there is C *such that, for* $t \in (0, T_1]$ *and* $x = X(\alpha,\lambda,t) \in S_t$*,*

$$\Big| D(x,t) - \frac{F_0(\alpha)}{|\partial_\alpha z_0(\alpha)|}\, n_0(\alpha) - \delta(\alpha)\,\tau_0(\alpha)\Big| \leq Ct.$$

In particular, if $\|M(\cdot,t)\|_{C^0} \leq C_1 t^2$ *and* $C_\Psi \equiv 0$*, the conclusion holds with* $\delta \equiv 0$*.*

Proof. Differentiating (4.10) and using (4.2),

$$\begin{aligned} t\kappa\,\nabla\Psi\cdot n_0 &= \partial_\lambda\big(\Psi\circ X\big) = \tfrac12(\Psi^+ - \Psi^-) = \tfrac12\Big(\int_0^{\alpha} M(\beta,t)\,\mathrm{d}\beta + C_\Psi(t)\Big), \\ \nabla\Psi\cdot\partial_\alpha X &= \partial_\alpha\big(\Psi\circ X\big) = \tfrac12(1-\lambda)F^- + \tfrac12(1+\lambda)F^+. \end{aligned}$$

By (4.13), $\nabla\Psi\cdot n_0 = -\delta + O(t)$ (we use $\min\kappa > 0$) and the right-hand side of the second identity is $F_0 + O(t)$. Since $\partial_\alpha X = |\partial_\alpha z_0|\tau_0 + O(t)$ and $\nabla\Psi$ is bounded, $\nabla\Psi\cdot\tau_0 = F_0/|\partial_\alpha z_0| + O(t)$. As $D = \nabla^\perp\Psi = (\nabla\Psi\cdot\tau_0)\,n_0 - (\nabla\Psi\cdot n_0)\,\tau_0$, the claim follows. □

Remark 4.6. Lemma 4.5 isolates the two components of the relaxed flux D in the strip. The normal component is fixed by the boundary fluxes $F^{\pm}$, i.e. by the Rankine–Hugoniot conditions; the tangential component $D\cdot\tau_0$ is the transverse difference quotient $-(\Psi^+ - \Psi^-)/(2t\kappa)$, and it is of order one exactly when the primitive of the flux mismatch is of order t. Making $M = O(t^2)$, as in [LM26], forces $D\cdot\tau_0 = O(t)$; prescribing instead the $O(t)$ part of M produces any desired tilt δ.

5. Proof of Theorem 1.1

We keep the notation of Section 4. The proof consists in choosing the normal acceleration a, the vertical correction h and the constant C_Ψ so that the flux mismatch has a prescribed first order part, $M = -2t\,\partial_\alpha(\kappa\delta) + O(t^2)$ with $\int_{\mathbb{T}} M = 0$, for a tilt δ to be chosen; and then in checking the open condition $D\cdot(D - 2a) < 0$ in the strip by comparison with its limit as $t \to 0$.

5.1. The limit $t \to 0$: velocity matching.

For the computation of the limits it is convenient to allow arbitrary normal velocities at $t = 0$.

Lemma 5.1. *Let* $z^{\pm}(\cdot,t)$ *be polynomials in* t *with coefficients in* $C^{3,\gamma}$*, with* $z^{\pm}(\cdot,0) = z_0$ *and* $\partial_t z^{\pm}(\cdot,0)\cdot n_0 = W^{\pm} \in C^{3,\gamma}(\mathbb{T})$*, such that the conclusions of Lemma 4.1 hold, and define* $f^{\pm}, v, F^{\pm}, M$ *by* (4.4)–(4.8)*. Let*

$$\overline{W} := \tfrac12\big[(1-\rho_{\mathrm{m}})W^+ + (1+\rho_{\mathrm{m}})W^-\big], \qquad P := 2\mathcal{K}_0[n_{0,2}] + 2A\mathcal{K}_0\overline{W}.$$

Then, uniformly on $\mathbb{T}$ *as* $t \to 0$*,* $v_S^{\pm}(\cdot,t) \to v_S^{(0)} := \mathcal{C}[z_0; f_0^+]^- + \mathcal{C}[z_0; f_0^-]^+$*, where* $f_0^{\pm} = [\![\rho]\!]^{\pm}(-e_2 + (n_{0,2} + AW^{\pm})n_0)$*, and*

$$v_S^{(0)}\cdot n_0 = P + \tfrac{A}{2}\big[(1+\rho_{\mathrm{m}})W^- - (1-\rho_{\mathrm{m}})W^+\big], \qquad v_S^{(0)}\cdot\tau_0 = \mathcal{T}_0\big[2(n_{0,2} + A\overline{W})\big] - \rho_{\mathrm{m}}\tau_{0,2}, \tag{5.1}$$

$$M(\cdot,t) \to M_0 := -\frac{4|\partial_\alpha z_0|}{1-\rho_{\mathrm{m}}^2}\Big[\big(I - 2A\mathcal{K}_0\big)\overline{W} - 2\mathcal{K}_0[n_{0,2}]\Big]. \tag{5.2}$$

For the ansatz (4.1)*, i.e.* $W^{\pm} = U_0 \pm c^{\pm}$*, we have* $\overline{W} = U_0$*,* $P = U_0$*,* $M_0 = 0$*,*

$$\begin{aligned} v_S^{(0)}\cdot n_0 = \mu_{\mathrm{m}}U_0 - A(1-\rho_{\mathrm{m}}^2)\kappa, \qquad v_S^{(0)}\cdot\tau_0 = -\mathcal{T}_0\sigma_0 - \rho_{\mathrm{m}}\tau_{0,2}, \\ F^{\pm}(\cdot,t) \to F_0 := |\partial_\alpha z_0|\big(-\sigma_0 - 2(1 - A\rho_{\mathrm{m}})\kappa\big). \end{aligned} \tag{5.3}$$

Proof. The convergence of $v_S^{\pm}$ follows from Lemma A.2: for instance $\mathcal{C}[z^+;f^+]^- \to \mathcal{C}[z_0;f_0^+]^-$, and $\mathcal{C}[z^-;f^-](z^+(\alpha,t)) \to \mathcal{C}[z_0;f_0^-]^+(\alpha)$ because $z^+(\alpha,t)$ lies above $\Gamma^-(t)$ and tends to $z_0(\alpha)$. By Lemma 2.3(b), $v_S^{(0)} = \langle \mathcal{C}[z_0;f_0^+ + f_0^-]\rangle - \frac{1}{2}f_0^+ + \frac{1}{2}f_0^-$, and $f_0^+ + f_0^- = -2e_2 + 2(n_{0,2} + A\overline{W})n_0$ because $[\![\rho]\!]^+ + [\![\rho]\!]^- = 2$. Since $\langle \mathcal{C}[z_0;e_2]\rangle = 0$ by Lemma 2.3(c), $\langle \mathcal{C}[z_0;f_0^+ + f_0^-]\rangle \cdot n_0 = P$ and $\langle \mathcal{C}[z_0;f_0^+ + f_0^-]\rangle \cdot \tau_0 = \mathcal{T}_0[2(n_{0,2} + A\overline{W})]$, while $f_0^{\pm}\cdot n_0 = A[\![\rho]\!]^{\pm}W^{\pm}$ and $f_0^{\pm}\cdot\tau_0 = -[\![\rho]\!]^{\pm}\tau_{0,2}$; this gives the normal part of (5.1) and $v_S^{(0)}\cdot\tau_0 = \mathcal{T}_0[2(n_{0,2}+A\overline{W})] - \rho_{\mathrm{m}}\tau_{0,2}$. Passing to the limit in (4.6)–(4.7),

$$\frac{M_0}{|\partial_\alpha z_0|} = \frac{2}{1-\rho_{\mathrm{m}}^2}\Big\{2v_S^{(0)}\cdot n_0 - (1-\rho_{\mathrm{m}})(1-A)W^+ - (1+\rho_{\mathrm{m}})(1+A)W^-\Big\} = \frac{4}{1-\rho_{\mathrm{m}}^2}\big(P-\overline{W}\big),$$

which is (5.2). For the ansatz, $(1-\rho_{\mathrm{m}})c^+ = (1+\rho_{\mathrm{m}})c^- = (1-\rho_{\mathrm{m}}^2)\kappa$ gives $\overline{W} = U_0$. Applying $\mathcal{K}_0$ to $(I - 2A\mathcal{K}_0)\sigma_0 = -2n_{0,2}$ and using $U_0 = -\mathcal{K}_0\sigma_0$ we get $(I-2A\mathcal{K}_0)U_0 = 2\mathcal{K}_0[n_{0,2}]$, i.e. $P = U_0$ and $M_0 = 0$; moreover $2(n_{0,2} + AU_0) = -\sigma_0$ by Proposition 2.7, which gives the tangential part of (5.3). The normal part gives $v_S^{(0)}\cdot n_0 = U_0 + A\rho_{\mathrm{m}}U_0 - A(1-\rho_{\mathrm{m}}^2)\kappa$, and inserting this in (4.6),

$$\frac{F_0}{|\partial_\alpha z_0|} = \frac{2}{1+\rho_{\mathrm{m}}}\Big(A(1+\rho_{\mathrm{m}})U_0 - (1+\rho_{\mathrm{m}})\kappa\big[A(1-\rho_{\mathrm{m}})+1-A\big]\Big) + 2n_{0,2} = -\sigma_0 - 2(1-A\rho_{\mathrm{m}})\kappa. \qquad \square$$

The vanishing of M at order t^0 is the statement that the centre of mass Z of the strip starts with the normal velocity of the Muskat problem, which we built into the ansatz.

5.2. The relaxation condition at $t = 0$.

Lemma 5.2. *Let $a^{(0)} := e_2 + \frac{A}{\mu_{\mathrm{m}}}v_S^{(0)}$ and, for a function δ, $D_\delta^{(0)} := \frac{F_0}{|\partial_\alpha z_0|}n_0 + \delta\,\tau_0$. Then*

$$a^{(0)}\cdot n_0 = -\frac{\sigma_0}{2} - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}, \qquad a^{(0)}\cdot\tau_0 = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}, \tag{5.4}$$

and, with Φ as in (1.5),

$$D_\delta^{(0)}\cdot\big(D_\delta^{(0)} - 2a^{(0)}\big) = \frac{2(1-A^2)}{\mu_{\mathrm{m}}}\,\Phi(\kappa) + \big(\delta - a^{(0)}\cdot\tau_0\big)^2. \tag{5.5}$$

In particular, for the tilt $\delta_* := a^{(0)}\cdot\tau_0 = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}$, the left-hand side equals $\frac{2(1-A^2)}{\mu_{\mathrm{m}}}\Phi(\kappa)$, which is negative under (1.6); and δ_* is the optimal tilt.

Proof. By (5.3), $a^{(0)}\cdot n_0 = n_{0,2} + AU_0 - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}$ with $n_{0,2} + AU_0 = -\sigma_0/2$; and $a^{(0)}\cdot\tau_0 = \tau_{0,2} - \frac{A}{\mu_{\mathrm{m}}}(\mathcal{T}_0\sigma_0 + \rho_{\mathrm{m}}\tau_{0,2}) = \frac{\tau_{0,2} - A\mathcal{T}_0\sigma_0}{\mu_{\mathrm{m}}}$, which equals $-\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}$ by (2.8). Writing $D_n = -\sigma_0 - 2(1-A\rho_{\mathrm{m}})\kappa$, $a_n = a^{(0)}\cdot n_0$ and $a_\tau = a^{(0)}\cdot\tau_0$, we have

$$D_n - 2a_n = -2\kappa\Big[(1-A\rho_{\mathrm{m}}) - \frac{A^2(1-\rho_{\mathrm{m}}^2)}{1+A\rho_{\mathrm{m}}}\Big] = -\frac{2\kappa(1-A^2)}{\mu_{\mathrm{m}}},$$

so $D_n(D_n - 2a_n) = \frac{2(1-A^2)}{\mu_{\mathrm{m}}}\kappa\big(2(1-A\rho_{\mathrm{m}})\kappa + \sigma_0\big)$. Adding $\delta(\delta - 2a_\tau) = (\delta - a_\tau)^2 - a_\tau^2$ and $a_\tau^2 = \frac{(1-A^2)^2\gamma_0^2}{4\mu_{\mathrm{m}}^2}$ gives (5.5). $\square$

Remark 5.3. Without tilt ($\delta = 0$) the condition $D_0^{(0)}\cdot(D_0^{(0)} - 2a^{(0)}) < 0$ reads $0 < 2(1-A\rho_{\mathrm{m}})\kappa < -\sigma_0$, the range of [LM26] for $\rho_{\mathrm{m}} = A$: it ignores the vortex sheet. The tilt δ_* moves $D^{(0)}$ onto the line through the centre $a^{(0)}$ of the relaxation disc parallel to n_0, and the admissible range becomes $\Phi(\kappa) < 0$. When $\gamma_0 \neq 0$ this range is strictly larger, and it is non-empty even where $\sigma_0 \geq 0$.

5.3. **Time regularity and acceleration matching.** The following lemma is proved in Appendix A.

Lemma 5.4. *Let $a \in C^{3,\gamma}(\mathbb{T})$ and let h be a polynomial with $h(0) = h'(0) = 0$. There are $T_1 > 0$ and C such that $F^\pm$ and M, extended to $t = 0$ by their limits, belong to $C^1([0,T_1]; C^0(\mathbb{T}))$, $\|\partial_t F^\pm(\cdot,t) - \partial_t F^\pm(\cdot,s)\|_{C^0} \le C|t-s|$, and*

$$\sup_{x = X(\alpha,\lambda,t) \in S_t} \big|v(x,t) - v_S^{(0)}(\alpha)\big| \le Ct^\gamma \qquad (0 < t \le T_1). \tag{5.6}$$

Lemma 5.5. *There is $G \in C^{3,\gamma}(\mathbb{T})$, depending only on $z_0, \kappa, \rho_{\mathrm{m}}$ and A, such that for every $a \in C^{3,\gamma}(\mathbb{T})$ and $h \equiv 0$,*

$$\partial_t M(\cdot,0) = G - \frac{4|\partial_\alpha z_0|}{1-\rho_{\mathrm{m}}^2}\big(I - 2A\mathcal{K}_0\big)a.$$

Proof. For fixed t the quantities $V^\pm$, $f^\pm$, v, $v_S^\pm$, $F^\pm$ and M are affine functions of the velocities $(\partial_t z^+, \partial_t z^-)$, with coefficients depending only on the positions $z^\pm(\cdot,t)$: $V^\pm$ is linear in $\partial_t z^\pm$, $f^\pm$ is affine in $V^\pm$, and v is linear in $f^\pm$. Write accordingly $M(t) = \mathcal{L}_t[\partial_t z^+, \partial_t z^-] + \mathcal{B}_t$. By the formulas of Appendix A, $\mathcal{L}_t$ and $\mathcal{B}_t$ are differentiable at $t = 0$ and their derivatives only involve z_0 and $\partial_t z^\pm(\cdot,0) = (U_0 \pm c^\pm)n_0$, which do not depend on a (the term $\frac{t^2}{2}an_0$ in the positions does not contribute at $t = 0$). Hence $\partial_t M(\cdot,0) = \mathcal{L}_0[\partial_t^2 z^+, \partial_t^2 z^-](\cdot,0) + G$ with G independent of a. Since $\partial_t^2 z^\pm(\cdot,0) = an_0$ and, by (5.2), the linear part of the map $(W^+, W^-) \mapsto M_0$ is $-\frac{4|\partial_\alpha z_0|}{1-\rho_{\mathrm{m}}^2}(I - 2A\mathcal{K}_0)\overline{W}$, we obtain the formula. The regularity $G \in C^{3,\gamma}$ is proved in Appendix A. □

The vanishing of M at order t^0 leaves no freedom (it is the Muskat equation for the velocity of the centre of mass), but at order t^1 the acceleration a is a free function, and by Proposition 2.6 we can make $\partial_t M(\cdot,0)$ equal to *any* function in $C^{3,\gamma}(\mathbb{T})$. This is the freedom used for the tilt.

Lemma 5.6 (Prescribed first order mismatch and mass correction)**.** *Let $\delta \in C^{4,\gamma}(\mathbb{T})$ and*

$$a := \big(I - 2A\mathcal{K}_0\big)^{-1}\Big[\frac{(1-\rho_{\mathrm{m}}^2)\big(G + 2\partial_\alpha(\kappa\delta)\big)}{4|\partial_\alpha z_0|}\Big] \in C^{3,\gamma}(\mathbb{T}), \tag{5.7}$$

so that $\partial_t M(\cdot,0) = -2\partial_\alpha(\kappa\delta)$ when $h \equiv 0$. Let $Q^{(0)}$ be the function Q of Lemma 4.3 computed with $h \equiv 0$, and $h(t) := -\frac{1}{4\pi}\int_0^t Q^{(0)}(s)\,\mathrm{d}s$. Then h is a polynomial with $|h(t)| \le Ct^3$, and the curves (4.1) with these a and h satisfy (4.12) and

$$\big\|M(\cdot,t) + 2t\,\partial_\alpha(\kappa\delta)\big\|_{C^0(\mathbb{T})} \le Ct^2 \qquad \text{for } t \in (0,T_1].$$

Proof. The choice (5.7) is possible by Proposition 2.6 with $k = 6$, $j = 3$, and Lemma 5.5 gives $\partial_t M(\cdot,0) = -2\partial_\alpha(\kappa\delta)$ for $h \equiv 0$. Let $M^{(0)}$ be the mismatch for $h \equiv 0$. By Lemmas 5.1 and 5.4, $M^{(0)}(\cdot,0) = 0$ and $\partial_t M^{(0)}$ is Lipschitz in time, so $\|M^{(0)}(\cdot,t) + 2t\partial_\alpha(\kappa\delta)\|_{C^0} \le Ct^2$. Since $\int_{\mathbb{T}} \partial_\alpha(\kappa\delta) = 0$, Lemma 4.3 gives $|Q^{(0)}(t)| \le Ct^2$. The areas $\mathcal{A}^\pm$ are polynomials in t, hence so are $Q^{(0)}$ and h, and $h(0) = h'(0) = 0$, $|h'(t)| \le Ct^2$, $|h(t)| \le Ct^3$. Adding $h(t)e_2$ to both curves adds $2\pi h$ to $\mathcal{A}^\pm$, so $Q = Q^{(0)} + 4\pi h' = 0$ and (4.12) holds by Lemma 4.3. Finally, Cauchy fields commute with translations, $\mathcal{C}[z + he_2; f](x + he_2) = \mathcal{C}[z;f](x)$, so the new $F^\pm$ are obtained from the old ones by adding $h'e_2$ to the velocities; by the affine dependence on the velocities and Lemma 2.3(b), $\|M - M^{(0)}\|_{C^0} \le C|h'(t)| \le Ct^2$. □

5.4. **Conclusion.**

Proof of Theorem 1.1. Let $\delta := \delta_* = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}} \in C^{4,\gamma}(\mathbb{T})$ be the optimal tilt of Lemma 5.2, choose a and h by Lemma 5.6, and set $C_\Psi(t) := -2t\kappa(0)\delta(0)$ in (4.9). Consider the curves (4.1) and the triple $(\bar\rho, \bar u, \bar m)$ of (4.11).

By Lemma 5.6, (4.12) holds and

$$\int_0^\alpha M(\beta,t)\,\mathrm{d}\beta + C_\Psi(t) + 2t\kappa(\alpha)\delta(\alpha) = \int_0^\alpha \big(M + 2t\partial_\beta(\kappa\delta)\big)(\beta,t)\,\mathrm{d}\beta = O(t^2)$$

uniformly in $\alpha \in [0, 2\pi]$; by Lemma 5.4 and since $h'(0) = 0$, $\|F^{\pm}(\cdot, t) - F_0\|_{C^0} \le Ct$ with F_0 given by (5.3). Thus (4.13) holds, and Lemma 4.5 gives $|D - D_\delta^{(0)}(\alpha)| \le Ct$ in S_t. By (5.6), $|a - a^{(0)}(\alpha)| \le \frac{|A|}{\mu_{\mathrm{m}}} Ct^\gamma$ in S_t, where $a = e_2 + \frac{A}{\mu_{\mathrm{m}}} v$ is the vector (3.5) of the state in the strip. By Lemma 4.2, $(\bar\rho, \bar u, \bar m)$ is bounded, solves (3.2) with initial datum ρ_0, and lies in K outside the strip. Since all quantities are bounded, Lemma 5.2 gives, in S_t,

$$D \cdot (D - 2a) \le \frac{2(1 - A^2)}{\mu_{\mathrm{m}}} \max_{\mathbb{T}} \Phi(\kappa) + C' t^\gamma < 0$$

for $0 < t \le T$, provided $T \le T_1$ and $C' T^\gamma < -\frac{2(1-A^2)}{\mu_{\mathrm{m}}} \max_{\mathbb{T}} \Phi(\kappa)$; note that $\max_{\mathbb{T}} \Phi(\kappa) < 0$ by (1.6) and continuity. Hence, with $\Omega_{\mathrm{mix}} := \{(x, t) : 0 < t < T,\ x \in S_t\}$, $(\bar\rho, \bar u, \bar m)$ is continuous on Ω_{mix} with values in $\mathcal{U}_A$ (Proposition 3.2), i.e. a subsolution in the sense of Definition 3.4. Theorem 3.5 yields infinitely many weak solutions with $\rho = \bar\rho$ outside Ω_{mix}, mixing in Ω_{mix}, and a sequence converging weakly-$*$ to $\bar\rho$. Since $\Omega_{\mathrm{mix}}(t) = S_t$ is the strip between $\Gamma^{\pm}(t)$ and Z has the form (1.7), all the assertions of Theorem 1.1 follow. □

Remark 5.7. The proof works with any tilt $\delta \in C^{4,\gamma}$ for which $\frac{2(1-A^2)}{\mu_{\mathrm{m}}} \Phi(\kappa) + (\delta - \delta_*)^2 < 0$ on $\mathbb{T}$. In particular, if κ satisfies the untilted condition $0 < 2(1 - A\rho_{\mathrm{m}})\kappa < -\sigma_0$, then $\delta = 0$ is admissible and Theorem 1.1 reduces to the single-layer result of [LM26] (with $\rho_{\mathrm{m}} = A$) in our periodic Hölder setting.

6. Necessary conditions: proof of Theorem 1.2

We now show that the range (1.6) cannot be improved by subsolutions with a single intermediate density. We first fix the class of subsolutions; it contains the subsolutions of Section 4, and its asymptotically flat analogue contains the single-layer subsolutions of [LM26].

Definition 6.1. Let $z_0 \in \mathcal{Z}^{3,\gamma}$, $\rho_{\mathrm{m}} \in (-1, 1)$ and $T > 0$. A *single-strip subsolution with intermediate density* ρ_{m} *emanating from* z_0 is a subsolution $(\bar\rho, \bar u, \bar m)$ on $[0, T]$ with initial datum $\rho_0 = \mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}$ (Definition 3.4), together with curves $z^{\pm}(\cdot, t)$ and a map X, such that:

(S1) $t \mapsto z^{\pm}(\cdot, t) - (\cdot, 0)$ belongs to $C^0([0, T]; C^{1,\gamma}(\mathbb{T})) \cap C^1([0, T]; C^{0,\gamma}(\mathbb{T}))$, $\delta(z^{\pm}(\cdot, t))$ is bounded below, $z^{\pm}(\cdot, 0) = z_0$, and for $t > 0$ the curve $\Gamma^+(t)$ lies in $\Omega^+(z^-(\cdot, t))$; we let $S_t := \Omega^-(z^+(\cdot, t)) \cap \Omega^+(z^-(\cdot, t))$;
(S2) $X(\cdot, \cdot, t) : \mathbb{T} \times [-1, 1] \to \overline{S_t}$ is C^1, $X(\cdot, \pm 1, t) = z^{\pm}(\cdot, t)$, and $|\partial_\lambda X| \le C_X t$;
(S3) $\bar\rho = 1$ in $\Omega^+(z^+(\cdot, t))$, $\bar\rho = \rho_{\mathrm{m}}$ in S_t, $\bar\rho = -1$ in $\Omega^-(z^-(\cdot, t))$, and $\Omega_{\mathrm{mix}} = \{(x, t) : 0 < t < T,\ x \in S_t\}$;
(S4) for every $t \in (0, T]$ the restrictions of $(\bar u, \bar m)(\cdot, t)$ to each of the three regions extend continuously to their closures;
(S5) $\bar u(x, t) \to 0$ as $|x_2| \to \infty$, uniformly in t.

The *initial normal velocities* of the strip are $W^{\pm} := \partial_t z^{\pm}(\cdot, 0) \cdot n_0 \in C^{0,\gamma}(\mathbb{T})$.

The subsolutions constructed in Section 4 satisfy (S1)–(S5), with X given by (4.2).

Proof of Theorem 1.2. Let $n^{\pm}$, $V^{\pm} = \partial_t z^{\pm} \cdot n^{\pm}$, $f^{\pm}$ and $v_S^{\pm}$ be defined from $z^{\pm}$ as in Section 4, and let $v := \bar u + A\bar m$ be the Darcy field of the subsolution.

Step 1: the Darcy field. By (3.2), $\operatorname{curl} v = -\partial_1 \bar\rho$ and $\operatorname{div} v = \operatorname{div} \bar u + A \operatorname{div} \bar m = -A \partial_t \bar\rho$ in $\mathcal{D}'$. By (S1) and (S3), $\partial_t \bar\rho = -\sum_{\pm} [\![\rho]\!]^{\pm} V^{\pm} \mathcal{H}^1 \llcorner \Gamma^{\pm}(t)$, so for a.e. t the field $v(\cdot, t)$ has the same distributional curl and divergence as $W := \mathcal{C}[z^+; f^+] + \mathcal{C}[z^-; f^-]$ (Lemma 2.3(b) and the computation of the jumps of $f^{\pm}$ in the proof of Lemma 4.2). Hence $w := v - W$ is a bounded, periodic field with $\operatorname{div} w = \operatorname{curl} w = 0$ in $\mathcal{D}'(\mathbb{R}^2)$; by Weyl's lemma $w^\flat$ is entire, and by Liouville's theorem on the cylinder it is constant. Outside the strip $\bar m = \pm \bar u$, so (S5) gives $v \to 0$ as $|x_2| \to \infty$, whereas $W \to \pm \frac{AQ}{4\pi} e_2$ (see the proof of Lemma 4.3). Comparing the limits at $\pm\infty$, the constant is 0 (and $AQ = 0$). Thus $v(\cdot, t) = W$ for a.e. t: the Darcy field of any single-strip subsolution is the one of our ansatz.

Step 2: boundary fluxes. By (S4), the conservation law $\partial_t\bar\rho + \operatorname{div}\bar m = 0$ and $\operatorname{div}\bar u = 0$ imply the Rankine–Hugoniot conditions $[\![\bar\rho]\!]V^\pm = [\![\bar m\cdot n^\pm]\!]$ and $[\![\bar u\cdot n^\pm]\!] = 0$ pointwise on $\Gamma^\pm(t)$. Inside the strip, $\operatorname{div}\bar m = 0$ and $\operatorname{div} v = 0$, so the field $D := \frac{2[(1-A^2)\bar m - (\rho_{\mathrm m}-A)v]}{1-\rho_{\mathrm m}^2} + 2e_2$ of (3.5) is divergence-free in S_t and continuous on $\overline{S_t}$. Reading the computation in the proof of Lemma 4.2 backwards, the Rankine–Hugoniot conditions determine the normal traces of D from inside the strip:

$$D\cdot n^\pm\,|\partial_\alpha z^\pm| = F^\pm \qquad \text{on } \Gamma^\pm(t), \tag{6.1}$$

with $F^\pm$ given by (4.6)–(4.7). Since $(\bar\rho,\bar u,\bar m)\in\mathcal{U}_A$ in S_t, D lies in the disc $\{|D-a|<|a|\}$, so $|D|\le 2|a|\le C$, with C independent of t because v is bounded uniformly in t (Lemma 2.3(b)).

Step 3: flux balance. On the periodic lift of $\overline{S_t}$, a simply connected strip, write $D = \nabla^\perp\Psi$ with $\Psi\in C^1$. By (6.1), $\partial_\alpha\big(\Psi\circ z^\pm\big) = F^\pm$, hence

$$\Psi(z^+(\alpha,t)) - \Psi(z^-(\alpha,t)) = \Psi(z^+(0,t)) - \Psi(z^-(0,t)) + \int_0^\alpha M(\beta,t)\,\mathrm d\beta.$$

By (S2) the left-hand side, and the first two terms on the right, are bounded by $\sup|D|\cdot 2C_X t \le Ct$. Thus $\big|\int_{\alpha_1}^{\alpha_2} M(\beta,t)\,\mathrm d\beta\big| \le Ct$ for all α_1,α_2.

Step 4: velocity of the centre of mass. By (S1) and Lemma A.2, $v_S^\pm(\cdot,t)\to v_S^{(0)}$ and $F^\pm(\cdot,t)\to F_0^\pm$ uniformly as $t\to 0$, where the limits are computed exactly as in the proof of Lemma 5.1 (which only uses the values of $z^\pm$ and $\partial_t z^\pm$ at $t=0$). In particular $M(\cdot,t)\to M_0$ uniformly, with M_0 given by (5.2). Step 3 gives $\int_{\alpha_1}^{\alpha_2} M_0 = 0$ for all α_1,α_2, so $M_0\equiv 0$, i.e. $(I-2A\mathcal{K}_0)\overline{W} = 2\mathcal{K}_0[n_{0,2}]$. Since U_0 solves the same equation (proof of Lemma 5.1) and $I-2A\mathcal{K}_0$ is injective (Proposition 2.6), $\overline{W} = U_0$. This is (a): writing $\kappa = \frac12(W^+ - W^-)$, the identity $(1-\rho_{\mathrm m})(W^+ - U_0) + (1+\rho_{\mathrm m})(W^- - U_0) = 0$ gives $W^\pm = U_0 \pm (1\pm\rho_{\mathrm m})\kappa$. Finally $\kappa\ge 0$: to first order in t, $\Gamma^\pm(t)$ is the normal graph $\{z_0(\alpha) + tW^\pm(\alpha)n_0(\alpha)\}$ over z_0, up to $o(t)$, and $\Gamma^+(t)$ lies above $\Gamma^-(t)$.

Step 5: the relaxation inequality. Fix α and $t\in(0,T]$. The trace of D at $z^+(\alpha,t)$ from inside the strip lies in the closed disc centred at $a_S^+ := e_2 + \frac{A}{\mu_{\mathrm m}}v_S^+(\alpha,t)$ with radius $|a_S^+|$, and by (6.1) its component along n^+ is $F^+/|\partial_\alpha z^+|$. Therefore

$$\Big(\frac{F^+}{|\partial_\alpha z^+|} - a_S^+\cdot n^+\Big)^2 \le |a_S^+|^2.$$

Let $t\to 0$. By Step 4 and part (a), the computation of Lemma 5.1 applies with $W^\pm = U_0\pm(1\pm\rho_{\mathrm m})\kappa$, so $F^+/|\partial_\alpha z^+| \to -\sigma_0 - 2(1-A\rho_{\mathrm m})\kappa$, and $a_S^+\to a^{(0)}$, whose components are given by (5.4). Writing D_n, a_n, a_τ as in the proof of Lemma 5.2, the limit inequality reads $D_n(D_n-2a_n) - a_\tau^2 \le 0$, and the computation there shows that the left-hand side equals $\frac{2(1-A^2)}{\mu_{\mathrm m}}\Phi(\kappa)$. Hence $\Phi(\kappa)\le 0$, which is (b). Since $\Phi(\kappa)>0$ for $\kappa>\kappa_{\max}$, we get $\kappa\le\kappa_{\max}$, and $\kappa_{\max}=0$ on $\Pi(z_0)$. □

Remark 6.2. The proof shows more precisely where the constraints come from. Part (a) is a consequence of mass conservation in a strip of width $O(t)$ (Step 3); part (b) only uses the relaxation inequality on the *boundary* of the strip, where the normal component of the relaxed flux is fixed by the Rankine–Hugoniot conditions. The tangential component is free in (b), and the bound $\Phi(\kappa)\le 0$ is attained by the optimal tilt of Lemma 5.2. Thus Theorems 1.1 and 1.2 together identify the admissible opening functions of single-strip subsolutions up to the boundary case $\Phi(\kappa)=0$.

7. Consequences: partially stable data, equal viscosities and viscous fingering

7.1. **Pinch-free data.**

Proof of Corollary 1.3. If $\Pi(z_0)=\emptyset$, then $\kappa_{\max}>0$ on $\mathbb{T}$ and $\kappa_{\max}\in C^{4,\gamma}(\mathbb{T})$ (end of Section 2), so $\kappa := \theta\kappa_{\max}$, $\theta\in(0,1)$, satisfies (1.6), and Theorem 1.1 applies. Conversely, if a single-strip subsolution with opening $\kappa>0$ everywhere exists, Theorem 1.2 gives $\kappa\le\kappa_{\max}$, hence $\kappa_{\max}>0$

everywhere, i.e. $\Pi(z_0) = \emptyset$. On $\{\sigma_0 \geq 0\}$ the untilted condition $2(1 - A\rho_{\mathrm{m}})\kappa < -\sigma_0$ fails for every $\kappa > 0$, so there the admissible opening comes entirely from the term $(1 - A^2)\gamma_0^2/(8\mu_{\mathrm{m}})$ in Φ, i.e. from the vortex sheet. □

The pinch set is the curved-interface version of the pinch singularity of [Men22]: by (5.4), $a^{(0)} = -\frac{\sigma_0}{2} n_0 - \frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}\tau_0 + O(\kappa)$, and the relaxation disc $\{|D - a| < |a|\}$ collapses as $\kappa \to 0$ exactly where $\sigma_0 \geq 0$ and $\gamma_0 = 0$. Note also that in the horizontally periodic setting the data are never fully stable: integrating (2.5) gives $\int_{\mathbb{T}} \sigma_0 \, \mathrm{d}s = -4\pi$ (Proposition 2.7), so $\sigma_0 < 0$ somewhere.

7.2. **Equal viscosities.**

Proof of Corollary 1.4. For $A = 0$, Example 2.12 gives $\sigma_0 = -2n_{0,2}$ and $\gamma_0 = -2\tau_{0,2}$, so $\Pi(z_0) = \{\tau_0 = -e_1\}$ and, since $n_{0,2}^2 + \tau_{0,2}^2 = 1$,

$$\kappa_{\max} = \frac{2n_{0,2} + \sqrt{4n_{0,2}^2 + 4\tau_{0,2}^2}}{4} = \frac{1 + n_{0,2}}{2}.$$

The result follows from Corollary 1.3 and Theorem 1.1; the regularity of $\kappa_{\max}$ is clear. □

For $A = 0$ the untilted range is $0 < \kappa < n_{0,2}$, which only allows mixing where the interface is a graph. The tilted range $0 < \kappa < \frac{1}{2}(1 + n_{0,2})$ coincides with it at horizontal points ($n_{0,2} = 1$, total opening speed 2, the maximal speed of [Ott99, Sze12]) and is strictly larger elsewhere. On the overturned part ($n_{0,2} < 0$) the tangential component of gravity drives the heavy fluid along the interface, the interface carries a vortex sheet $\gamma_0 = -2\tau_{0,2} \neq 0$, and the relaxation allows a mixing zone of normal width up to $(1 + n_{0,2})t$. Mixing is only impossible where the interface is horizontal with the heavy fluid below. In [CFM22], mixing zones for turned interfaces and bubbles are instead localised near the unstable region; a closed bubble always has a point with $n_0 = -e_2$, where by Theorem 1.2 a single strip cannot open.

7.3. **Viscous fingering driven by injection.**

The original experiments of Saffman and Taylor [ST58] concern a less viscous fluid injected into a more viscous one, without gravity. Consider the problem with gravity constant $G \geq 0$ and a uniform velocity Ve_2 at infinity:

(7.1)
$$\partial_t \rho + \operatorname{div}(\rho u) = 0, \qquad \operatorname{div} u = 0, \qquad (1 + A\rho)u = -\nabla p - G\rho\, e_2, \qquad u(x,t) \to Ve_2 \ \text{ as } |x_2| \to \infty.$$

Corollary 7.1. *Let $G \geq 0$, $A \in (-1, 1)$ and $V \in \mathbb{R}$ with $G_{\mathrm{eff}} := G + AV > 0$. Let $z_0 \in \mathcal{Z}^{6,\gamma}$, $\rho_{\mathrm{m}} \in (-1, 1)$ and $\kappa \in C^{4,\gamma}(\mathbb{T})$ with $0 < \kappa < \kappa_{\max}$, where $\kappa_{\max}$ and $\Pi(z_0)$ are computed from z_0 and A as in (1.5) (they do not depend on G and V). Let T, Z and $z^{\pm}$ be given by Theorem 1.1. Then there are infinitely many weak solutions of (7.1) on $[0, T/G_{\mathrm{eff}}]$ with initial datum $\mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}$ which are pure outside, and mixing inside, the strip between the curves*

$$z^{\pm}\big(\cdot, G_{\mathrm{eff}}\, t\big) + Vt\, e_2.$$

In particular, for $G = 0$ and $AV > 0$ (the displacing fluid is the less viscous one), mixing solutions exist for all pinch-free data.

Proof. Let $(\tilde\rho, \tilde u)$ be one of the solutions given by Theorem 1.1, and set

$$\rho(x,t) := \tilde\rho\big(x - Vte_2,\, G_{\mathrm{eff}} t\big), \qquad u(x,t) := Ve_2 + G_{\mathrm{eff}}\, \tilde u\big(x - Vte_2,\, G_{\mathrm{eff}} t\big).$$

Then $\operatorname{div} u = 0$, and $\partial_t \rho + u \cdot \nabla \rho = G_{\mathrm{eff}}\big(\partial_t \tilde\rho + \tilde u \cdot \nabla \tilde\rho\big) = 0$ in the weak sense. Moreover, $(1 + A\rho)u + G\rho e_2 = (1 + A\rho)Ve_2 + G\rho e_2 + G_{\mathrm{eff}}(1 + A\rho)\tilde u = Ve_2 + G_{\mathrm{eff}}\big((1 + A\tilde\rho)\tilde u + \tilde\rho e_2\big)$, whose curl vanishes because $\operatorname{curl}((1 + A\tilde\rho)\tilde u) = -\partial_1 \tilde\rho$. Hence $(1 + A\rho)u = -\nabla p - G\rho e_2$ for some pressure, and $u \to Ve_2$ at infinity. The mixing properties are invariant under this change of variables. □

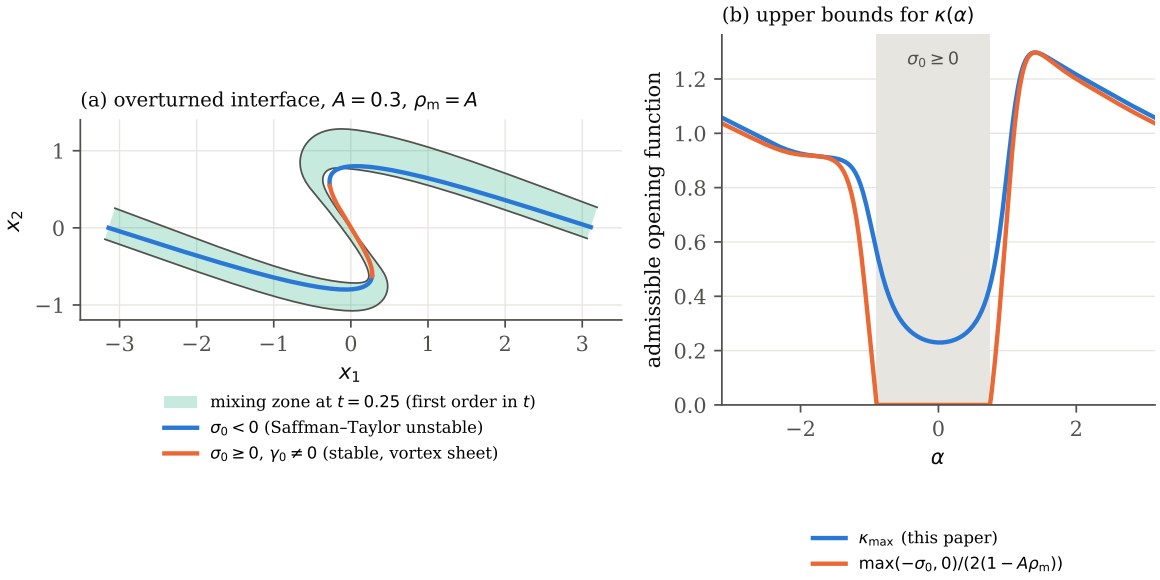


FIGURE 1. Overturned interface $z_0(\alpha) = (\alpha - 1.5\sin\alpha, 0.8\sin\alpha)$ with $A = 0.3$ and $\rho_m = A$. (a) The interface, coloured by the sign of the Saffman–Taylor function σ_0, and the strip between $z_0 + t(U_0 \pm (1 \pm \rho_m)\kappa)n_0$ at $t = 0.25$ for $\kappa = 0.8\,\kappa_{max}$ (the first order part of the mixing zone of Theorem 1.1). The mixing zone surrounds the whole interface, including the stable part. (b) The maximal opening function κ_{max} of Theorem 1.1 compared with the untilted bound $\max(-\sigma_0, 0)/(2(1 - A\rho_m))$; the shaded region is $\{\sigma_0 \geq 0\}$, where the untilted bound vanishes.

Thus injection enters only through the effective gravity $G_{eff} = G + AV$: the viscous contribution AV to the Saffman–Taylor function of the flat interface adds to the gravitational one, and in the frame moving with the injection the problem is the one studied above. For $G = 0$ the instability is purely viscous, and the vortex-sheet term γ_0 in κ_{max} comes entirely from the viscosity contrast.

7.4. **An example.** Let $z_0(\alpha) = (\alpha - 1.5\sin\alpha,\ 0.8\sin\alpha)$. Then $\partial_\alpha z_0 = (1 - 1.5\cos\alpha,\ 0.8\cos\alpha) \neq 0$, and z_0 is injective modulo $2\pi e_1$: if $z_{0,2}(\alpha) = z_{0,2}(\beta)$ with $\beta \not\equiv \alpha$ then $\beta \equiv \pi - \alpha \pmod{2\pi}$, and $z_{0,1}(\alpha) - z_{0,1}(\pi - \alpha) = 2\alpha - \pi$ lies in $2\pi\mathbb{Z}$ only if $\alpha \equiv \frac{\pi}{2} \pmod{\pi}$, in which case $\beta \equiv \alpha$. Hence $z_0 \in \mathcal{Z}^{k,\gamma}$ for every k. The interface is overturned: $n_{0,2} < 0$ for $\cos\alpha > 2/3$. The tangent is horizontal only where $\cos\alpha = 0$, and there $\tau_0 = e_1$. By Corollary 1.4, for $A = 0$ the data are pinch-free although Rayleigh–Taylor stable on about 27% of the interface, and mixing solutions exist with any opening $0 < \kappa < \frac{1}{2}(1 + n_{0,2})$.

For $A \neq 0$ we computed σ_0 and γ_0 numerically, solving the second kind equations (1.4) with the trapezoidal rule on 512 nodes (the kernel of $\mathcal{K}_0$ is smooth). We checked the identity (2.8), with $\mathcal{T}_0$ evaluated by the alternating-point rule, to a relative accuracy of 10^{-13}. For $A \in \{0.3, -0.6, 0.9\}$ the Saffman–Taylor function is non-negative on 16% to 26% of the interface, while $|\gamma_0| \geq 1.7$ there; so the data are pinch-free and $\kappa_{max} \geq 0.22$ on $\mathbb{T}$ (with $\rho_m = A$). Figure 1 shows the case $A = 0.3$. The scripts are in the `checks/` folder of the source repository.

8. Remarks and open problems

8.1. **Continuous profiles with tilt (formal).** Single strips do not exhaust the relaxation. Consider a mixing zone of width $O(t)$ around Z in which the density has a continuous profile $\bar\rho = R(\xi)$, $\xi = (x - Z)\cdot n/t$, increasing from -1 at ξ_- to 1 at ξ_+ and normalised so that Z

is the centre of mass. Repeating the computations of Section 5 at leading order gives, with $\mathcal{P}(\xi) := -\int_{\xi_-}^{\xi} \eta R'(\eta)\,\mathrm{d}\eta \geq 0$,

$$D\cdot n = -\sigma - \frac{2(1-AR)\,\mathcal{P}}{1-R^2}, \qquad a\cdot n = -\frac{\sigma}{2} - \frac{A^2\mathcal{P}}{1+AR}, \qquad a\cdot\tau = -\frac{(1-A^2)\gamma}{2(1+AR)},$$

where for the last identity we used that, at leading order, the tangential component of the Darcy field is $v\cdot\tau = -\mathcal{T}_0\sigma - \tau_2 R$ across the zone, together with (2.8). Without tilt the condition is $(1-AR)\mathcal{P} < -\frac{\sigma}{2}(1-R^2)$, and its extremal profile has fronts at $\xi_+ = -\sigma/(1-A)$ and $\xi_- = \sigma/(1+A)$; these are the opening speeds reached by the multilayer subsolutions of [LM26, Section 7.5] as the number of layers tends to infinity, and for the flat interface they give Example 3.6 with $\theta \to 1$. With the optimal tilt the condition becomes $(D\cdot n - a\cdot n)^2 < |a|^2$ pointwise in the zone, which again enlarges the admissible profiles wherever $\gamma \neq 0$. We expect that multilayer subsolutions with a tilt in every layer realise these larger profiles, by the argument of Section 5 applied layer by layer; note that each layer carries its own stream function, so one tilt per layer is available, while the number of compatibility conditions per order in t grows accordingly.

8.2. **The role of the vortex sheet.** Theorem 1.2 shows that within single strips the vortex sheet is not an additional assumption but a structural part of the relaxation: the admissible opening is governed by $|a|$, which combines $-\frac{1}{2}\sigma_0$ and $-\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}$. This is a porous-media counterpart of the vortex-sheet driven mixing for the Euler equations [MS23]. Physically, on a Saffman–Taylor stable but inclined part of the interface, the heavy fluid slides down along the interface, and the relaxation allows this shear to produce a mixing zone of positive width. Whether such mixing zones are selected by a physical criterion, for instance Otto's relaxation [Ott99] or maximal dissipation, is an interesting question; for $A = 0$ and the flat interface the maximal mixing rate is selected [Ott99, Sze12].

8.3. **Open problems.**

(1) *Pinch points.* When $\Pi(z_0) \neq \emptyset$, Theorem 1.2 forces $\kappa = 0$ on $\Pi(z_0)$, and a mixing zone must be localised, as in [CFM22] for $A = 0$. Extending Theorem 1.1 to opening functions vanishing on $\Pi(z_0)$ requires weighted versions of the estimates of Lemma 4.5, since the margin in (5.5) degenerates there.
(2) *Asymptotically flat interfaces.* Our periodic setting simplifies the mass balance (Lemma 4.3); the asymptotically flat setting of [LM26] controls the primitive of M by L^1-type estimates instead. The tilt requires the primitive of the first order part $-2\partial_\alpha(\kappa\delta)$ to be bounded, which holds when $\kappa\delta - \kappa_\infty\delta_\infty$ decays.
(3) *Longer times and lower regularity.* The time expansion only gives a short existence time, and we assumed $z_0 \in C^{6,\gamma}$, far from the H^5 of [LM26] and the $C^{3,\alpha}$ of [FS18] for $A = 0$.
(4) *Exact relaxation.* We only used that $\mathcal{U}_A$ consists of laminates; Luna-Velasco and Mengual [LM26] prove that its closure is the full Λ-convex hull and the weak relaxation of (1.1). Our set $\mathcal{U}_A$ coincides with their set O: with $g(\rho,u,m) := (1-A\rho)|m-\rho u|^2 + (1-\rho^2)(m-\rho u)\cdot(Au+e_2)$, a direct computation gives $D\cdot(D-2a) = \frac{4(1-A^2)}{(1-\rho^2)^2(1+A\rho)}\,g$. Hence Theorem 1.2 cannot be improved by enlarging the target set of the subsolutions.

Appendix A. Periodic Cauchy integrals

It is convenient to work with complex densities. For $z \in \mathcal{Z}^{1,\gamma}$ and $G \in C^{0,\gamma}(\mathbb{T};\mathbb{C})$ set

$$\mathfrak{c}[z;G](x) := \frac{1}{4\pi i}\int_{\mathbb{T}} G(\beta)\,\partial_\beta z(\beta)\cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta, \qquad x \in \mathbb{C}\setminus\Gamma,$$

so that $\mathcal{C}[z;f]^\flat = \mathfrak{c}[z;f^\flat]$.

A.1. **Reduction to closed curves.** With $w = e^{ix}$ and $\omega = e^{i\zeta}$, a direct computation gives

$$\tfrac{1}{2}\cot\tfrac{\zeta - x}{2}\,\mathrm{d}\zeta = \frac{\mathrm{d}\omega}{\omega - w} - \frac{i}{2}\,\mathrm{d}\zeta.$$

The map $x \mapsto e^{ix}$ sends the cylinder $\mathbb{T}\times\mathbb{R}$ to $\mathbb{C}\setminus\{0\}$, Γ to a closed Jordan curve γ of class $C^{k,\gamma}$ winding once around the origin, $\Omega^+(z)$ to the punctured interior of γ and $\Omega^-(z)$ to its exterior. Hence $\mathfrak{c}[z;G]$ is the classical Cauchy integral $\frac{1}{2\pi i}\int_\gamma \frac{\tilde G(\omega)\,\mathrm{d}\omega}{\omega - w}$ of the transported density $\tilde G$, minus the constant $\frac{1}{4\pi}\int_{\mathbb{T}} G\partial_\beta z\,\mathrm{d}\beta$. The classical Plemelj–Privalov theorem for closed $C^{1,\gamma}$ curves [Mus53, Chapter 2], [Kre14, Chapter 7] then gives Lemma 2.3(b): $\mathfrak{c}[z;G]$ extends to a $C^{0,\gamma}$ function on each closed region, with norm bounded by $C\|G\|_{C^{0,\gamma}}$, where C depends only on γ, $\delta(z)$ and $\|z\|_{\mathcal{Z}^{1,\gamma}}$; its boundary values are given by the Plemelj formulas $\mathfrak{c}^\pm = \pm\frac{1}{2}G + \mathrm{p.v.}\,\mathfrak{c}$. The value at $w = 0$ corresponds to the limit $x_2 \to +\infty$.

A.2. **Derivative identities.**

Lemma A.1. *Let $z \in \mathcal{Z}^{2,\gamma}$ and $G \in C^{1,\gamma}(\mathbb{T};\mathbb{C})$. Then, in $\mathbb{C}\setminus\Gamma$,*

$$\partial_x\,\mathfrak{c}[z;G] = \mathfrak{c}\Big[z;\frac{\partial_\beta G}{\partial_\beta z}\Big]. \tag{A.1}$$

If $z = z(\beta,t)$ and $G = G(\beta,t)$ are C^1 in t with values in $\mathcal{Z}^{2,\gamma}$ and $C^{1,\gamma}$, then

$$\partial_t\,\mathfrak{c}[z;G](x) = \mathfrak{c}\Big[z;\partial_t G - \frac{\partial_\beta G}{\partial_\beta z}\,\partial_t z\Big](x), \qquad x \notin \Gamma(t). \tag{A.2}$$

Proof. Since $\partial_\beta \cot\frac{z(\beta)-x}{2} = -\frac{1}{2}\csc^2\frac{z(\beta)-x}{2}\,\partial_\beta z$, we have $\partial_x \cot\frac{z(\beta)-x}{2} = -\partial_\beta \cot\frac{z(\beta)-x}{2}/\partial_\beta z$ and $\partial_t \cot\frac{z(\beta)-x}{2} = \partial_\beta \cot\frac{z(\beta)-x}{2}\cdot \partial_t z/\partial_\beta z$. Integrating by parts in β (there are no boundary terms by periodicity),

$$\partial_x\mathfrak{c} = -\frac{1}{4\pi i}\int G\,\partial_\beta\cot\,\mathrm{d}\beta = \frac{1}{4\pi i}\int \partial_\beta G\cot\,\mathrm{d}\beta,$$

which is (A.1), and

$$\partial_t\mathfrak{c} = \frac{1}{4\pi i}\int\Big[(\partial_t G\,\partial_\beta z + G\,\partial_\beta\partial_t z)\cot - \partial_\beta(G\partial_t z)\cot\Big]\,\mathrm{d}\beta = \frac{1}{4\pi i}\int\big(\partial_t G\,\partial_\beta z - \partial_\beta G\,\partial_t z\big)\cot\,\mathrm{d}\beta,$$

which is (A.2). □

Iterating (A.1) and using the Plemelj–Privalov theorem gives Lemma 2.3(d): if $z \in \mathcal{Z}^{j+1,\gamma}$ and $G \in C^{j,\gamma}$ then all derivatives of order $\le j$ of $\mathfrak{c}[z;G]$ are Cauchy integrals of $C^{0,\gamma}$ densities, hence $C^{0,\gamma}$ up to Γ. If $x = x(t)$ is a C^1 path in $\Omega^\pm(z(t))$, then since $\mathfrak{c}$ is holomorphic,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathfrak{c}[z;G]\big(x(t)\big) = \mathfrak{c}\Big[z;\partial_t G - \frac{\partial_\beta G}{\partial_\beta z}\partial_t z\Big](x) + \mathfrak{c}\Big[z;\frac{\partial_\beta G}{\partial_\beta z}\Big](x)\,\dot x, \tag{A.3}$$

and by uniform convergence of the derivatives up to Γ the same formula holds for the boundary values $t \mapsto \mathfrak{c}[z;G]^\pm(\alpha)$ with $x(t) = z(\alpha,t)$.

A.3. **Continuity.**

Lemma A.2. *Let $z_k \to z$ in $\mathcal{Z}^{1,\gamma}$ and $G_k \to G$ in $C^{0,\gamma}(\mathbb{T};\mathbb{C})$. Then $\mathfrak{c}[z_k;G_k]^\pm \to \mathfrak{c}[z;G]^\pm$ uniformly on $\mathbb{T}$, and if $x_k \in \overline{\Omega^+(z_k)}$ with $x_k \to z(\alpha)$, then $\mathfrak{c}[z_k;G_k](x_k) \to \mathfrak{c}[z;G]^+(\alpha)$; similarly from below.*

Proof. By the Plemelj–Privalov theorem the functions $\mathfrak{c}_k := \mathfrak{c}[z_k;G_k]$ are uniformly $C^{0,\gamma}$ on $\overline{\Omega^+(z_k)}$, with a constant L independent of k. Given $\varepsilon > 0$ let $y = z(\alpha) + \varepsilon n(\alpha)$; for k large $y \in \Omega^+(z_k)$ at distance $\ge \varepsilon/2$ from Γ_k, so $\mathfrak{c}_k(y) \to \mathfrak{c}[z;G](y)$ by dominated convergence. Hence

$$\limsup_k\,|\mathfrak{c}_k(x_k) - \mathfrak{c}[z;G]^+(\alpha)| \le \limsup_k\,\big(L|x_k - y|^\gamma + L\varepsilon^\gamma\big) \le 2L\varepsilon^\gamma,$$

and ε is arbitrary. Uniformity in α follows from the same argument and the equicontinuity. □

A.4. **Proof of Lemma 5.4 and of the regularity of G in Lemma 5.5.** By Lemma 4.1, the curves $z^\pm(\cdot,t)$ are polynomials in t with coefficients in $C^{3,\gamma}$, with $\delta(z^\pm) \geq \frac{1}{2}\delta(z_0)$ on $[0,T_1]$. The normals $n^\pm$, the lengths $|\partial_\alpha z^\pm|$ and the velocities $V^\pm = \partial_t z^\pm \cdot n^\pm$ are therefore smooth in t with values in $C^{2,\gamma}$, and so are the complex densities $G^\pm := (f^\pm)^\flat$ and their time derivatives.

By (4.5), $(v_S^+)^\flat = \mathfrak{c}[z^+;G^+]^- + \mathfrak{c}[z^-;G^-]\circ z^+$, where $z^+(\alpha,t) \in \Omega^+(z^-(\cdot,t))$ for $t>0$, and similarly for v_S^-. Applying (A.3) once, $\partial_t v_S^\pm$ is a sum of boundary values and evaluations on the other curve of Cauchy integrals with densities

$$G_1^\pm = \partial_t G^\pm - \frac{\partial_\beta G^\pm}{\partial_\beta z^\pm}\partial_t z^\pm, \qquad G_2^\pm = \frac{\partial_\beta G^\pm}{\partial_\beta z^\pm},$$

multiplied by $\partial_t z^\pm$; these densities are bounded in $C^{1,\gamma}$ uniformly in t. Applying (A.3) once more, $\partial_t^2 v_S^\pm$ involves Cauchy integrals of densities bounded in $C^{0,\gamma}$, evaluated on $\Gamma^\pm(t)$ from the side of the strip, and these are bounded uniformly in $t\in(0,T_1]$ by the Plemelj–Privalov theorem. Since $F^\pm$ are polynomials in $v_S^\pm\cdot n^\pm$, $V^\pm$, $n_2^\pm$ and $|\partial_\alpha z^\pm|$, we obtain that $\partial_t F^\pm$ and $\partial_t^2 F^\pm$ are bounded in $C^0(\mathbb{T})$ on $(0,T_1]$. By Lemma A.2 applied to the densities appearing in $F^\pm$ and $\partial_t F^\pm$, these functions extend continuously to $t=0$, which proves the first part of Lemma 5.4.

For (5.6), let $x = X(\alpha,\lambda,t)$, so that $|x - z^\pm(\alpha,t)| \leq 2t\kappa(\alpha)$. The uniform Hölder bounds on $\overline{\Omega^-(z^+)}$ and $\overline{\Omega^+(z^-)}$ give $|v(x,t) - v_S^+(\alpha,t)| \leq Ct^\gamma$, and the Lipschitz bound in time just proved gives $|v_S^+(\alpha,t) - v_S^{(0)}(\alpha)| \leq Ct$.

Finally, $G = \partial_t M(\cdot,0)$ for $a=0$, $h=0$ is obtained by letting $t\to 0$ in the formula for $\partial_t M$. In the limit all Cauchy integrals become boundary values on $z_0 \in \mathcal{Z}^{6,\gamma}$, and the densities involve $n_0 \in C^{5,\gamma}$, $\partial_t z^\pm(\cdot,0) = (U_0 \pm c^\pm)n_0 \in C^{4,\gamma}$, their first α-derivatives, and $\partial_t^2 z^\pm(\cdot,0) = 0$. Thus $G^\pm(\cdot,0) \in C^{4,\gamma}$, $G_1^\pm(\cdot,0), G_2^\pm(\cdot,0) \in C^{3,\gamma}$, and by Lemma 2.3(d) the boundary values are in $C^{3,\gamma}$. The remaining factors ($\partial_t z^\pm(\cdot,0)$, n_0, and the time derivatives at $t=0$ of $n^\pm$, $V^\pm$ and $|\partial_\alpha z^\pm|$, which involve $\partial_\alpha\partial_t z^\pm(\cdot,0)$) belong to $C^{3,\gamma}$. Hence $G \in C^{3,\gamma}(\mathbb{T})$. □

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